

Wavesim Workbench

Electromagnetic simulation from FreeCAD geometry

Version 0.1

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Introduction

1.1 What the workbench does, in a nutshell

The Wavesim workbench turns FreeCAD solids into an electromagnetic simulation. You assign a material to each body, say where the simulated region ends, place sources and monitors, and press Run. The Wavesim solver performs the computation, and the results are returned to the FreeCAD document tree as plots and tables.

This manual is written for a physicist. It assumes knowledge of (among others): Maxwell's equations, complex impedance and Fourier transforms, and explains the numerical methods as far as is required to judge a result. Boxes marked **THE PHYSICS · SKIPPABLE** can be skipped without losing the thread. Red boxes mark common mistakes that give a wrong answer without an error. Chapter 4 works through two complete examples step by step, one for each solver.



Figure 1.1: The Wavesim toolbar, in its five groups. Chapter 5 describes every button.

1.2 Front end and solver: the mental model

The workbench (front end) and the Wavesim solver (back end) are separate programs that communicate through files on disk.

The workbench lives inside FreeCAD. It knows about geometry, materials, ports and monitors; it does *no* physics. The solver is its own project, installed separately. It knows nothing about FreeCAD, CAD geometry or the user interface. It is handed a grid, some material arrays and a list of what to record, and it solves Maxwell's equations.

Each is installed and configured separately; hence the two Python installations (Chapter 2) and the run folder left on disk (Chapter 11). This manual is about the front end: what you do in FreeCAD and how to use the workbench.

1.3 The solvers

The workbench currently offers two solvers, selected with **Solve:** in the Simulation panel. Both act on the same geometry, materials and mesh.

Full wave (FDTD), the default, solves Maxwell's equations in the time domain (Chapter 8). Excitation by a short pulse yields the response over a broad frequency band in a single run.

Electrostatic solves $\nabla \cdot (\epsilon \nabla \varphi) = 0$ for conductors held at fixed potentials or charges, and returns the potential, the field, the conductor charges and the capacitance matrix (Chapter 9).

Toolbar buttons that do not apply to the selected solver are greyed out (Section 5.2).

1.4 The workflow at a glance

Every simulation follows the same steps (Figure 1.2). Steps 1–3 are done once for a model. You will usually go round steps 4–8 several times, refining the mesh until the answer stops changing. The mesh is the step most likely to give a wrong answer without any warning: a transmission line, for example, needs more cells across its cross-section than its wavelength alone asks for. Chapter 4 shows this happening, and Chapter 6 explains the controls that fix it.

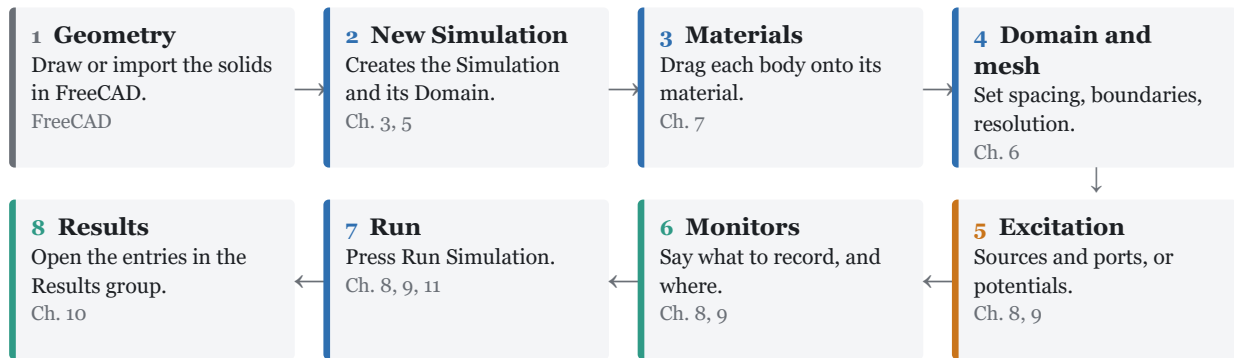


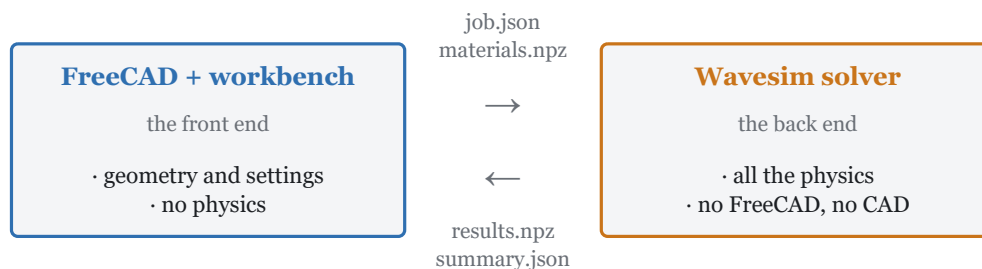
Figure 1.2: The workflow, and the chapter that covers each step.

Installation and setup

2.1 Requirements

- FreeCAD 1.1.
- The Wavesim solver repository: the folder that contains the `wavesim` package folder.
- A separate Python 3.10 or newer with `numpy`, `scipy` and `numba`. SPICE co-simulation additionally requires `PySpice` and the `ngspice` shared library.

A second Python is required because FreeCAD's bundled interpreter lacks the solver's numerical stack, while the solver's interpreter cannot open a FreeCAD document. The workbench is therefore given the path of the solver's interpreter (Section 2.3). The two processes share no memory and no imports: the workbench writes the job to a folder and launches the solver as a separate process, which writes its results back to the same folder (Figure 2.1; Chapter 11).



Two separate programs. They share files on disk and nothing else.

Figure 2.1: Two Pythons. The workbench runs inside FreeCAD's Python; the solver runs in its own.

2.2 Installation

Workbench. Copy the workbench folder into FreeCAD's `Mod` folder: `%APPDATA%\FreeCAD\v1-1\Mod\` on Windows, and elsewhere the `Mod` subfolder of the path returned by `FreeCAD.getUserAppDataDir()` in FreeCAD's Python console. **The folder must be named `wavesim-workbench`**, since the workbench locates its icons and solver launcher by that name. Restart FreeCAD.

Solver environment. Create a Python environment with the three packages, for example with `conda`:

```
conda create -n wavesim python=3.11 -y
conda install -n wavesim numpy scipy -y
conda run -n wavesim pip install numba
```

A `venv` with `pip` is equally suitable. The Wavesim repository may be placed anywhere on disk. It need not be installed into the environment, as the workbench adds it to the solver's search path at each run.

SPICE co-simulation (optional). Install `PySpice` into the same environment and obtain the `ngspice shared library` (`ngspice.dll` on Windows), not the console program. It is required only for the features in Appendix A.

2.3 The Settings dialog

The dialog opens from the `Settings...` button, the last on the Wavesim toolbar, or from the Wavesim menu. Its six rows are:

- **Solver Python interpreter:** the `python.exe` of the solver environment.

- **Wavesim repository:** the folder that *contains* the `wavesim` package folder, not the package folder itself.
- **Results output folder:** the runs of each document are written to a subfolder named after the document; by default beside FreeCAD's user data. Chapter 11 lists its contents.
- **ngspice.dll:** required only for SPICE ports.
- **Solver backend:** leave on Automatic. The CUDA GPU option is listed, but **vo.1 has no GPU support** (Section 11.3).
- **Voxelisation:** the number of processes used to slice the geometry before the solver starts.
Serial (one process) gives the same result, more slowly.

The settings are stored in `wavesim_settings.json` beside FreeCAD's user data (the dialog shows the full path), not in FreeCAD's preferences, and therefore survive a preferences reset. Changes take effect at the next run without restarting FreeCAD.

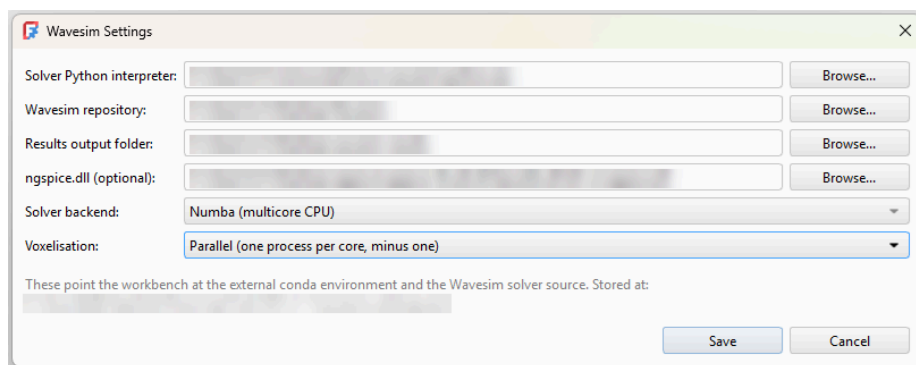


Figure 2.2: The Settings dialog.

2.4 Verifying the installation

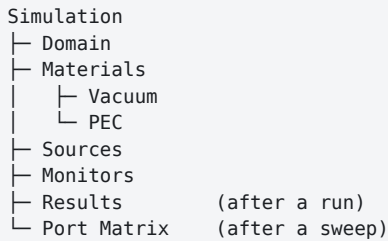
1. **The workbench loads.** Select **Wavesim** in FreeCAD's workbench selector. The Wavesim toolbar appears, with the cross-section toolbar below it. Otherwise, the Report view (View → Panels → Report view) contains a line beginning `Wavesim: failed to load`.
2. **The repository is found.** If FreeCAD was started before the repository was set, the Report view shows `Wavesim: solver path not found: .` The warning is harmless once the setting is saved.
3. **The solver environment is complete.** The following command prints nothing if all three packages are installed:
`<your solver environment>\python.exe -c "import numpy, scipy, numba"`
4. **A run completes.** Save a new document and press `New Simulation`, then `Run Simulation`. With no bodies assigned to a material, the workbench runs a small demo box and reports `Wavesim: no materials assigned; running the demo box`. The progress window first shows `Loading solver (first run may compile, please wait)...`; the first run of a session may pause here while the solver compiles. FreeCAD accepts no input other than the progress window until the run ends. On success, the Report view prints `Wavesim: run complete in`, followed by the run time and the output folder.

On failure, an error box gives the cause. `Solver Python interpreter not found:` indicates a wrong interpreter path. The solver run failed: followed by `ModuleNotFoundError` indicates a package missing from the solver environment, or a wrong repository path.

Concepts and the document tree

3.1 The Simulation container and its groups

All objects the workbench adds to a document are held under a single **Simulation** object in the model tree. **New Simulation** creates it with the layout below.



- **Domain** — the box that is solved, its boundaries and its mesh. Exactly one exists, created with the Simulation and always listed first.
- **Materials** — one entry per material, with the bodies made of it nested beneath.
- **Sources** — point sources, modal, lumped and SPICE line ports, and Gaussian beams.
- **Monitors** — everything that records during a run.
- **Results** — what the last run produced.
- **Port Matrix** — the output of a **Port Matrix (Z / Y)** sweep.

A document holds a single Simulation; **New Simulation** is greyed out once one exists. The Simulation is created with two materials, **Vacuum** and **PEC**, and with the Domain background — the medium of every cell no body fills — set to Vacuum.

Double-clicking **Simulation** opens its panel, which holds the settings that apply to the whole run: the solver (**Full wave (FDTD)** by default, or **Electrostatic**), the display units, the maximum simulation time (default **2 ns**) and the maximum frequency (default **1 GHz**).

Each run replaces the results of the previous one: **Run Simulation** deletes the Results group and builds a new one. The Port Matrix group sits beside Results rather than inside it, and is therefore left untouched by an ordinary run.

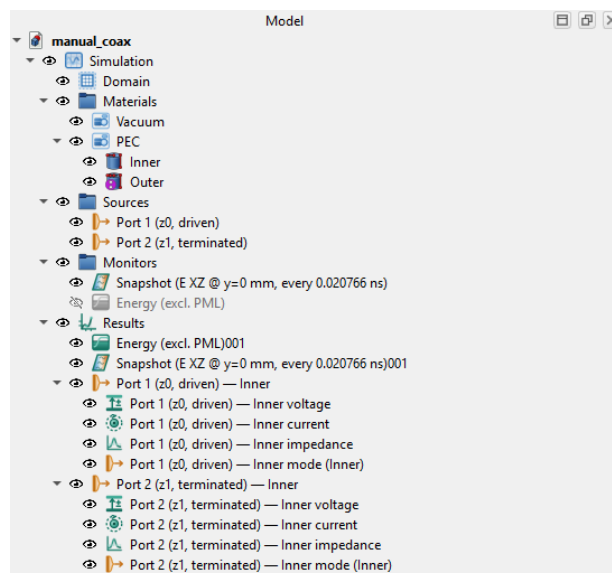


Figure 3.1: The model tree after a run.

3.2 Units: CAD length units and the solver's SI

The solver works in SI units (metres, seconds, hertz). The workbench converts on export, so an SI length is never entered.

Lengths. FreeCAD stores every length in millimetres and displays it in the property editor in the unit system selected in FreeCAD’s preferences. **The Wavesim panels always show lengths in millimetres**, irrespective of that setting.

Time and frequency. Each Simulation has its own display units, set in its panel: `s`, `ms`, `us`, `ns`, `ps`, `fs` for time (default **ns**) and `Hz`, `kHz`, `MHz`, `GHz`, `THz` for frequency (default **GHz**). They are used by every panel that shows a time or a frequency. Changing a unit re-expresses the same physical value — 1.5 GHz becomes 1500 MHz — and has no effect on the run.

Expressions are evaluated in seconds and hertz

Behind the panels, times are stored in **seconds** and frequencies in **hertz**. An expression bound to one of these properties (Section 3.4) is evaluated in those units, not in the unit the panel shows. An expression `8` on the maximum frequency means 8 Hz, not 8 GHz; write `8e9`. The property editor’s tooltip on each such property says which unit it is in.

3.3 Automatic domain sizing

The domain is not drawn. It is the smallest box that contains:

- every body assigned to a material;
- every point source, and both ends of every lumped port and SPICE line port;
- every voltage and current monitor path, and every snapshot plane;

plus a **background spacing** set per face. The box is empty until a body is assigned to a material, and is resized whenever a body is assigned, its geometry edited, or an object placed outside the current box.

A body not assigned to a material takes no part in the simulation: it is ignored by the domain and never reaches the solver. Chapter 6 covers the box, its boundaries and its mesh in full.

3.4 Parametric models: expressions and VarSets

Any property that shows the `f(x)` button in FreeCAD’s property editor can be bound to an expression — a formula, or a parameter kept in a **VarSet** (FreeCAD’s variable set object). A single parameter can thereby drive a sweep: bind several properties to the same VarSet parameter, change the parameter and run again.

The following properties accept expressions:

- on the Simulation: `MaxTime` and `MaxFrequency` ;
- on a material: `Eps`, `Mu` and `Sigma` ;
- on a source or port drive: the waveform parameters, such as `Fmax`, `Amplitude`, `Frequency`, `RiseTime` .

An expression owns its property: it is re-evaluated at every recompute, so a value typed into the panel would be discarded. A bound row is therefore **greyed out**, with the expression named in its tooltip:

Driven by the expression '...'. Edit it in the property editor (the `f(x)` button); a value typed here would be overwritten on the next recompute.

Each run replaces the results of the previous one (Chapter 11). Keeping the results of two parameter values side by side therefore requires a saved copy of the document for each.

Quick start: two worked examples

This chapter follows two short but complete paths through the workbench, one for each solver: a coaxial line solved in full wave, and a parallel-plate capacitor solved electrostatically. Each is worked through in the order of Figure 1.2, each step names the chapter that covers it in depth, and each result is checked against an analytic value. The finished documents are `manual/examples/manual_coax.FCStd` and `manual/examples/manual_capacitor.FCStd`.

4.1 A coaxial line

The first model is a 60 mm air-filled coaxial line, terminated at both ends by modal ports and driven with a pulse over 0–8 GHz.

A coaxial line is used for two reasons. Its cross-section carries a pure TEM mode whose characteristic impedance follows from the radii alone, so the workbench can be checked against an analytic value at two independent points: the mode solver, before any time stepping, and the impedance recovered from the full-wave run. And it is the geometry on which the default mesh fails, for a reason that applies to every transmission line (Section 4.1.3).

The results of this example will be compared to the analytic answer. For an air line of inner radius a and outer radius b ,

$$Z_0 = \frac{\eta_0}{2\pi} \ln(b/a) = 49.94 \quad \Omega \quad (b/a = 2.3).$$

4.1.1 The model

1. **Draw the geometry.** In FreeCAD's Part workbench, create a `Cylinder` of radius **3.0 mm** and height **60 mm**, and a `Tube` of inner radius **6.9 mm**, outer radius **11.0 mm** and the same height, both at the origin. Name them `Inner` and `Outer`. Save the document: the run folder is named after the file (Chapter 11).
2. **Create the simulation.** Switch to the Wavesim workbench and press `New Simulation`. The Simulation appears in the tree with its Domain, its groups and the materials **Vacuum** and **PEC** (Section 3.1).
3. **Assign both bodies to PEC.** Drag `Inner` and `Outer` in the model tree and drop them onto **PEC**, beneath which they are then listed. The domain sizes itself to the two solids (Section 3.3).
4. **Set the frequency band.** Double-click **Simulation** and set `Max frequency:` to **8 GHz**, leaving `Max simulation time:` at **2 ns** and the solver at `Full wave (FDTD)`. The panel reports the resulting `Time steps:`. Conformal (cut-cell) PEC conductors is on by default and is left on (Section 7.4).

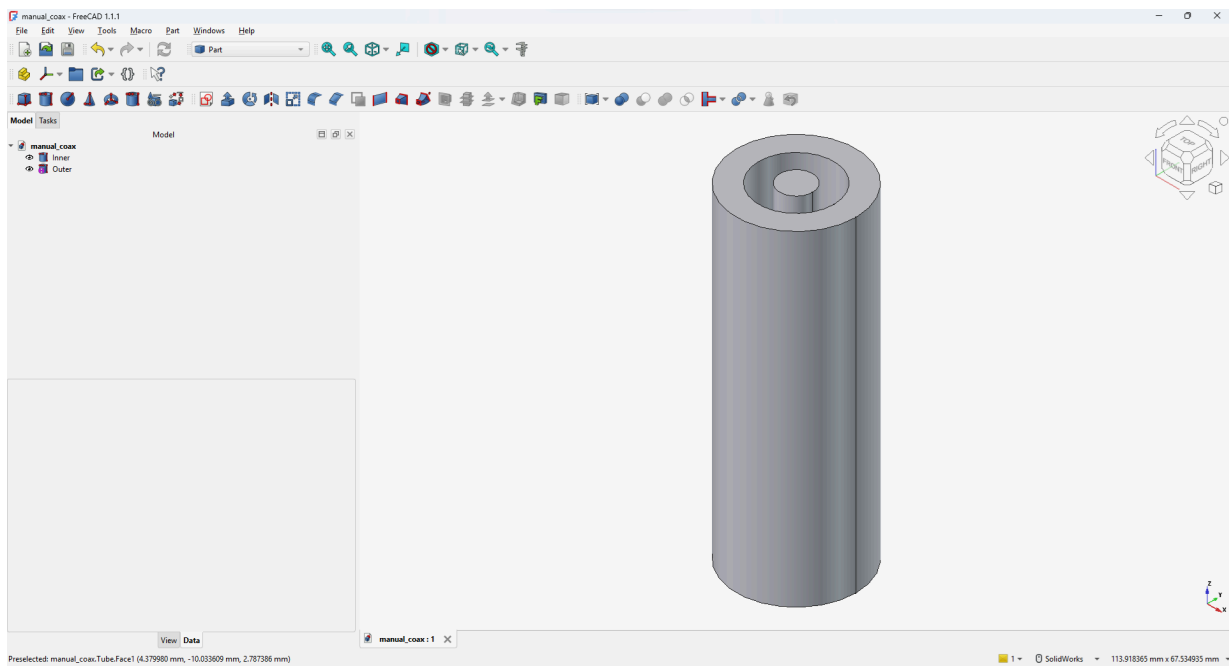


Figure 4.1: The two solids before any simulation object exists.

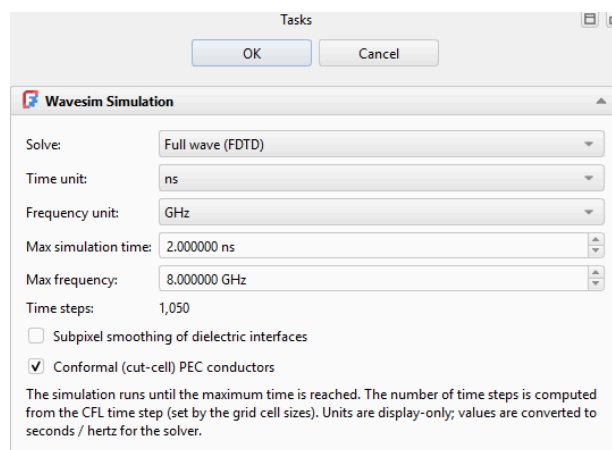


Figure 4.2: The Simulation panel, with the band set to 8 GHz.

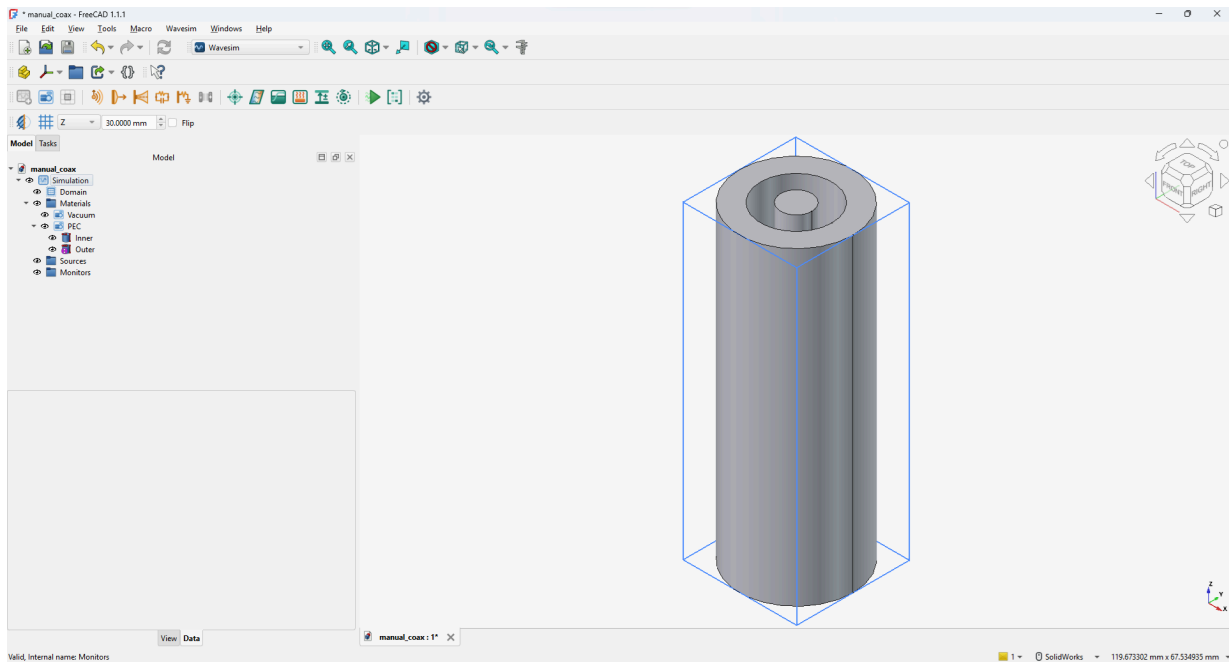


Figure 4.3: The tree after both bodies are assigned to PEC.

4.1.2 The ports

A **modal port** terminates a domain face with the mode of the PEC cross-section on that face. It launches that mode inward and absorbs what returns, so no absorbing boundary is needed on that face (Section 8.3).

1. **Add the driven port.** Press **Add Modal Port**. In its panel set **Port face:** to **Z min (z0)**, tick **Drive** on the **Inner** row and choose the **Gaussian Pulse** waveform at amplitude **1.0**. The panel names the reference conductor below the table: **Outer** touches the edge of the solve region, so it is ground and carries no mode of its own.
2. **Set the waveform's band.** Set the panel's own **Max frequency:** to **8 GHz**.
3. **Add the terminating port.** Press **Add Modal Port** again, set **Port face:** to **Z max (z1)** and leave **Drive** unticked. An undriven modal port is a pure absorber, and it still terminates its face.

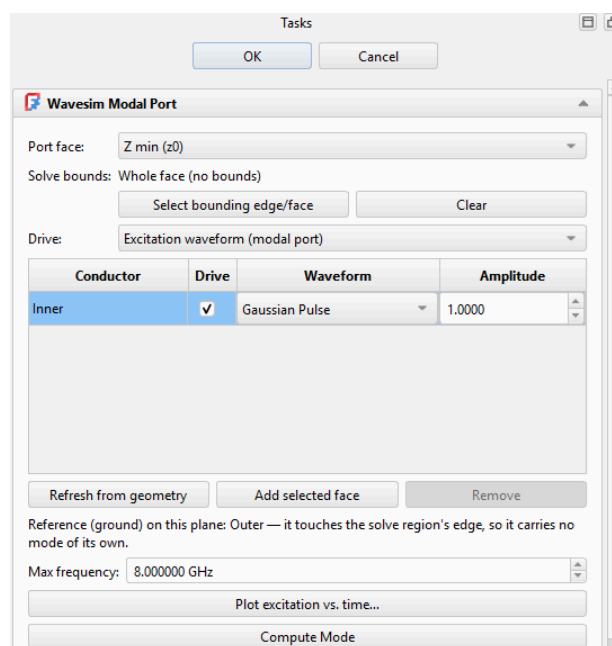


Figure 4.4: The Modal Port panel on the z-min face.

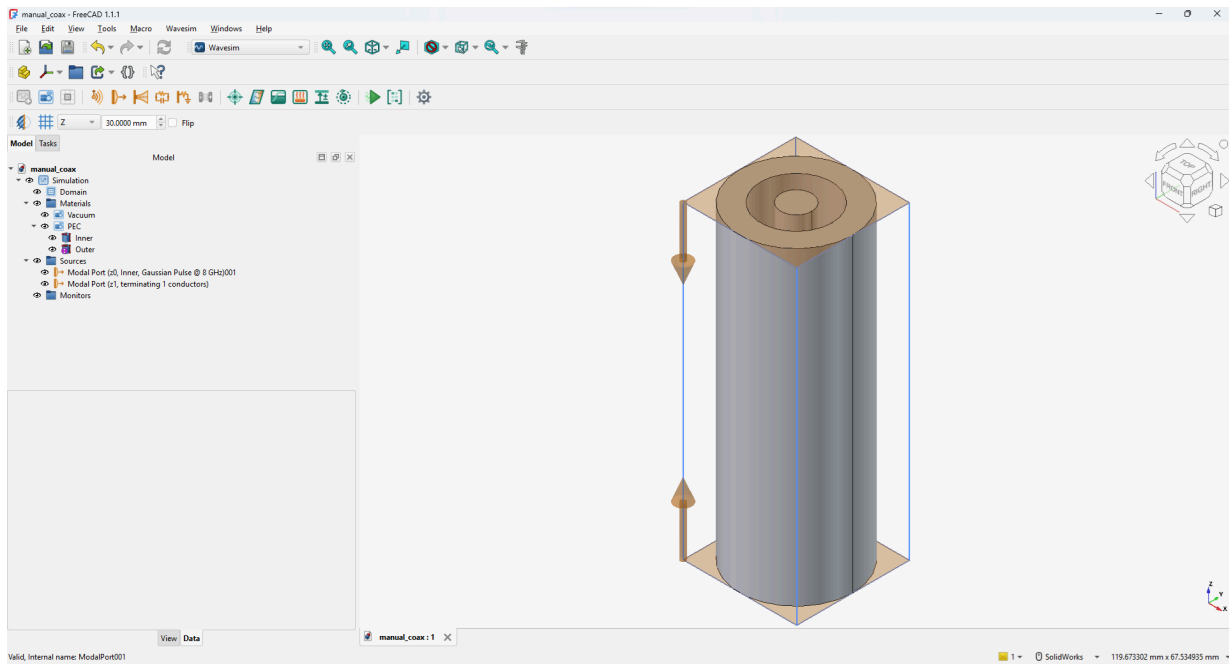


Figure 4.5: Both port planes, with the domain box.

4.1.3 The mesh

Press **Compute Mode** at the foot of the port panel. The mode solver solves the two-dimensional electrostatic problem on the port plane alone and reports Z_0 without an FDTD run, so a mesh can be assessed before the full simulation is run.

At the default **Resolution:** of 20 cells per wavelength it reports **52.87 Ω** , 6% above the analytic value. At 8 GHz the wavelength is 37 mm, which puts eight cells across the 6.9 mm bore (Figure 4.6). A transmission line is electrically small across its axis, so a mesh derived from the wavelength does not resolve the cross-section that carries the mode.

Select both bodies and press **Refine Body Mesh** and ask for **12** cells across **Inner** and **44** across **Outer**, which puts 0.5 mm cells across the bore. The request is a lower bound: the mesher also lays grid planes on the faces the body itself presents, and the bore wall of **Outer** is one of them, so the 44 requested across its span are delivered as 46 (Section 6.6). **Compute Mode** now reports **50.17 Ω** , within 0.5% of the analytic value.

The two mode pictures differ in more than smoothness. At 20 cells per wavelength the bore is an octagon four cells across, and the potential falls from the inner conductor to the shield in that many steps; the refined mesh resolves the circular boundary and the $1/r$ field between the conductors (Figure 4.7). Z_0 is an integral over that field, which is why a cross-section resolved to a few cells is already 6% out.

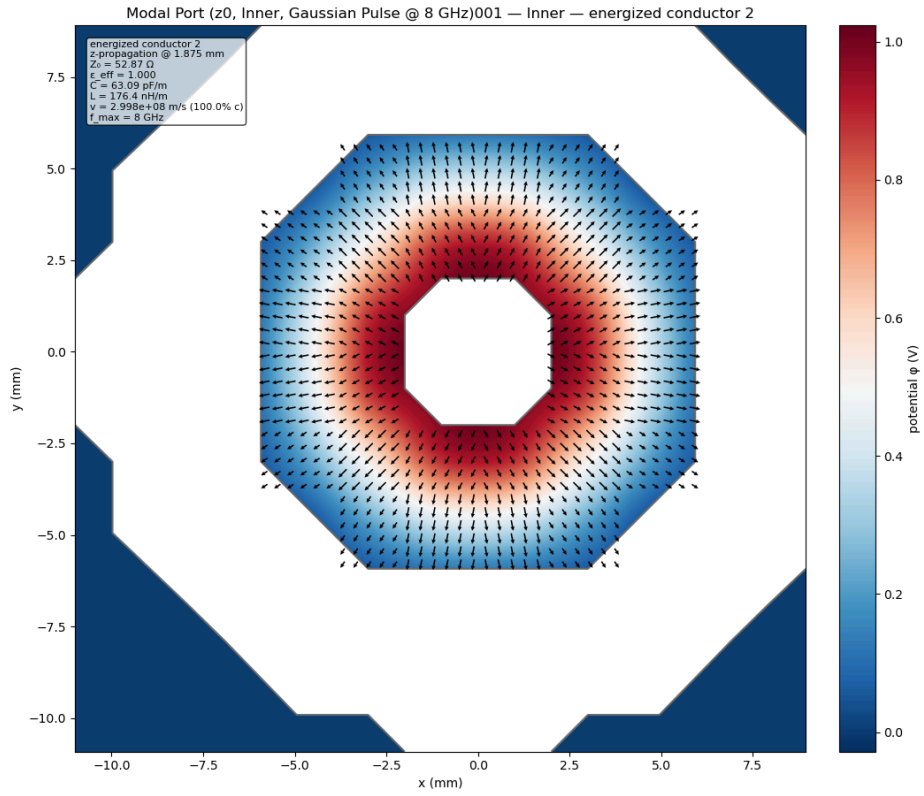


Figure 4.6: 20 cells/ λ : eight cells across the bore, $Z_0 = 52.87 \Omega$.

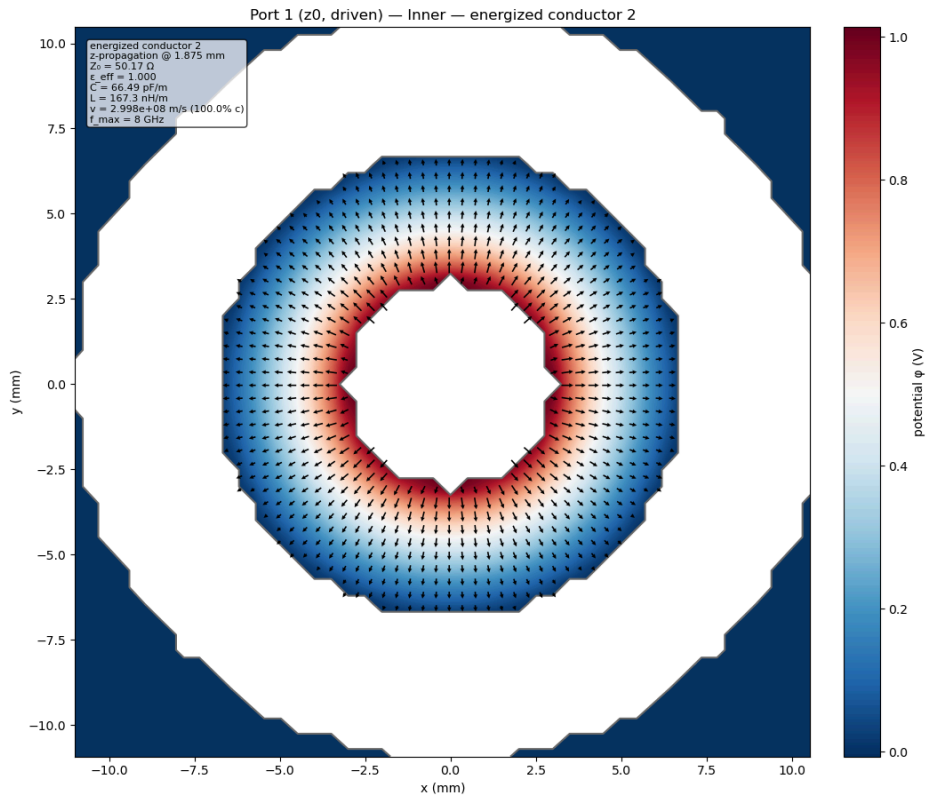


Figure 4.7: Refined: 0.5 mm cells, $Z_0 = 50.17 \Omega$.

Table 4.1: Z_0 against mesh, from Compute Mode .

Mesh	Grid	Cells	Z_0 (Ω)	Error
20 cells/ λ	$11 \times 11 \times 32$	3,872	52.87	+5.87%
40 cells/ λ	$22 \times 22 \times 64$	30,976	49.61	-0.66%
80 cells/ λ	$44 \times 44 \times 128$	247,808	49.93	-0.02%
20 cells/ λ + refinement	$46 \times 46 \times 32$	67,712	50.17	+0.46%

The first three rows converge to the analytic value, reaching 0.02% at 80 cells per wavelength. The refined mesh attains comparable accuracy with 3.7 times fewer cells, which are concentrated where the fields vary.

Refinement reduces the cell count without reducing the number of time steps: the time step follows the smallest cell anywhere in the grid, so it falls as soon as 0.5 mm cells exist. The refined mesh runs 1,927 steps against 2,181, 12% fewer, while the cell count falls by a factor of 3.7 (Section 6.8).

4.1.4 The run

1. **Add a snapshot monitor.** Press **Add Snapshot** , set **Field:** to **E** , **Slice plane:** to **XZ** at offset **0** and **Record every:** to about **0.02 ns**. The panel converts that to a step count and a frame count (Section 8.5).
2. **Add an energy monitor.** Press **Add Energy Monitor** and leave it on the interior of the domain. It records the total electromagnetic energy at every step, from which the decay of the excitation can be read afterwards.
3. **Run.** Press **Run Simulation** . FreeCAD is unresponsive while the solver runs (Chapter 11). The summary window (Figure 4.8) reports the grid, the time step, the step count and the wall time.
4. **Read the results.** The Results group holds one entry per monitor, and a group per port containing its mode shape, its voltage and current, and $Z(f)$ (Chapter 10).

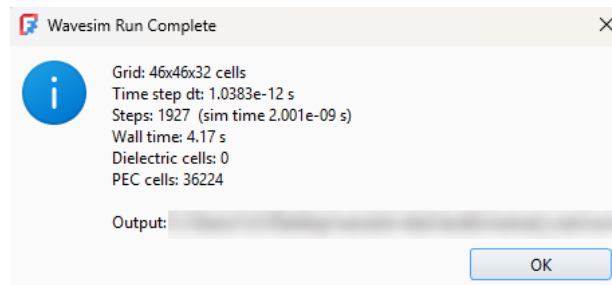


Figure 4.8: The run summary.

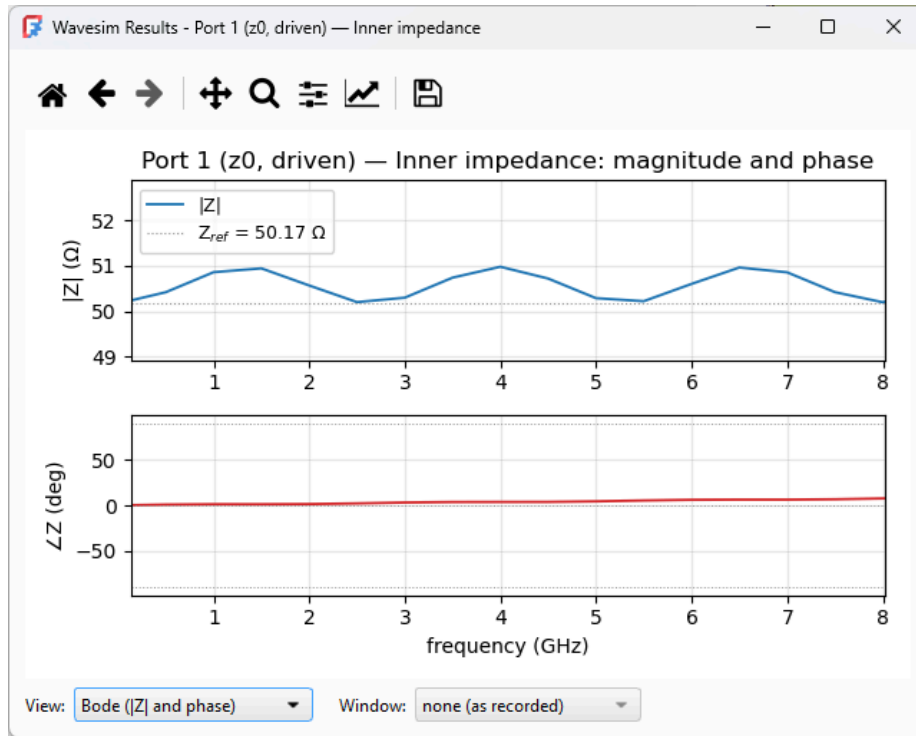


Figure 4.9: $Z(f)$ at port 1, Bode view.

The port's group in Results holds $Z(f)$, which is not a separate measurement: the port records the modal voltage and current on its plane at every step, and the entry is the ratio of their Fourier transforms over the band the pulse carried (Section 10.3). A single run therefore gives the impedance across 0–8 GHz.

The full-wave $|Z(f)|$ (Figure 4.9) lies between 50.1 and 51.0 Ω across that band, at a phase of a few degrees: the flat, real impedance of a matched line. It brackets the 50.17 Ω the mode solver reported on the same mesh, and sits 0.3–2% above the analytic 49.94 Ω . The ripple has a period of about 2.5 GHz = $c/(2 \times 60 \text{ mm})$, the round-trip resonance of the 60 mm line rather than numerical noise: the two port planes are not perfectly matched to each other, and what little they reflect interferes along the line.

A run always ends at the maximum simulation time, whatever the fields are doing; the energy monitor is what distinguishes a run that ended with the domain empty from one cut off while it still held energy. Here its entry falls from a peak of $7.7 \times 10^{-13} \text{ J}$, as the pulse enters, to $1.8 \times 10^{-23} \text{ J}$ at 2 ns — eleven orders of magnitude, so the pulse left through the ports and nothing is still ringing in the domain. An energy that levels off instead indicates a field still trapped in the domain, which truncates the transform and makes $Z(f)$ unreliable at the low end (Section 8.6).

Chapter 6 covers the mesh controls used here, Chapter 8 the ports and monitors, and Chapter 10 the result windows.

4.2 A parallel-plate capacitor

The second model is two square plates, each $10 \times 10 \times 1 \text{ mm}$, facing each other across a 1.0 mm vacuum gap centred on $z = 0$ (Figure 4.10). The textbook answer is

$$C = \epsilon_0 \frac{A}{d} = \frac{8.854 \times 10^{-12} \cdot 10^{-4}}{10^{-3}} = 0.885 \text{ pF}$$

That formula is a lower bound. It assumes the field stops at the plate edges; in fact the field bulges past them, and this fringing field stores extra charge. The size of that excess is what the solve provides and a hand calculation cannot.

4.2.1 The model

1. **Draw the geometry.** In a new document, create two Part boxes of **10 × 10 × 1 mm**: **Bottom** at $(-5, -5, -1.5)$ mm and **Top** at $(-5, -5, 0.5)$ mm. Save the document.
2. **Create the simulation.** Press **New Simulation**, double-click **Simulation** and set **Solve:** to **Electrostatic**. The panel hides the time and frequency rows and shows **Extract the capacitance matrix**, which is on by default (Section 9.2).
3. **Assign both plates to PEC.** Drag both boxes onto **PEC**. The cell size becomes 0.1 mm: the thinnest dimension of any body, the 1 mm plate thickness, divided by the default 10 cells per feature (Section 6.4).
4. **Clear the plates from the box.** In the Domain panel, set the background spacing to **3 mm** on all faces and leave every face on **Ground**. A plate touching a Ground face would be shorted to it (Section 9.4). This gives a $160 \times 160 \times 90$ grid of 2.3 M cells.

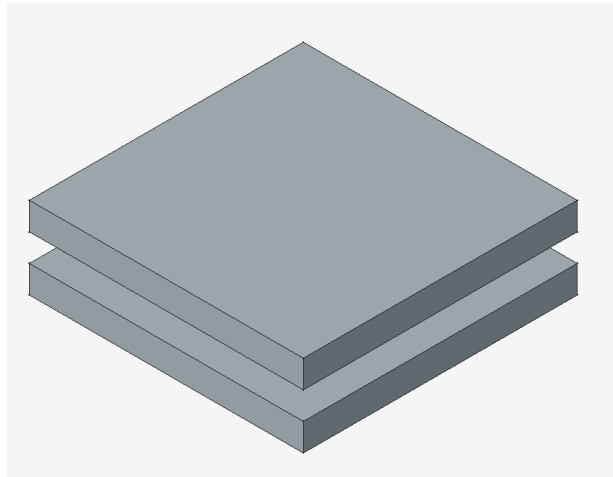


Figure 4.10: The two plates.

4.2.2 The potentials and monitors

1. **Set the potentials.** Press **Set Potential**, set **Top** to **Potential** at 1 V and **Bottom** to **Potential** at 0 V, and press OK (Section 9.3).
2. **Add two snapshot monitors.** Press **Add Snapshot** twice, both on the **XZ** plane at offset 0, one with **Field:** set to **phi** and the other to **E**.

4.2.3 The run

Press **Run Simulation**. The Results group gains the two slices and a **Capacitance matrix** entry, which opens the window of Figure 4.11 (Section 9.5).

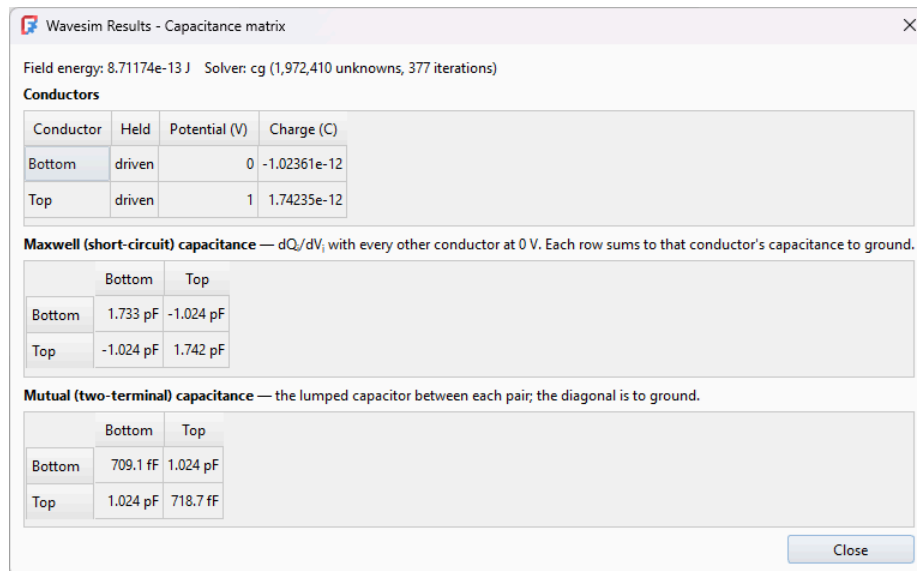


Figure 4.11: The capacitance window for the parallel-plate capacitor.

The window gives the **mutual** capacitance between the plates as **1.024 pF**, 16% above $\epsilon_0 A/d$. The Maxwell diagonal, 1.742 pF for **Top**, is not the capacitor: it adds the 0.72 pF between **Top** and the grounded box (Section 9.1).

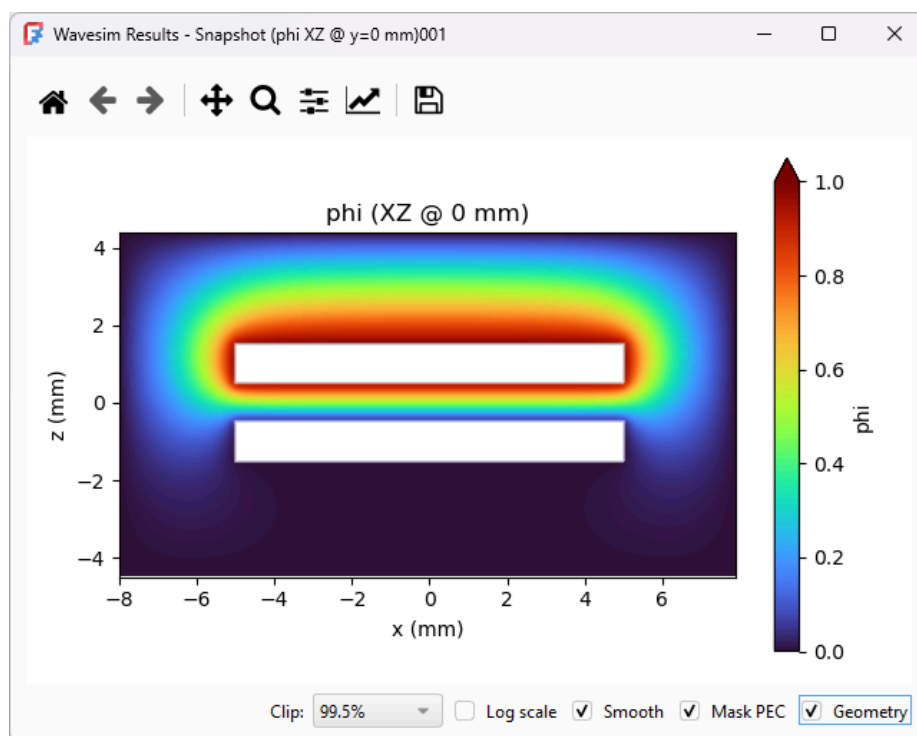


Figure 4.12: The potential ϕ on the XZ plane through the centre.

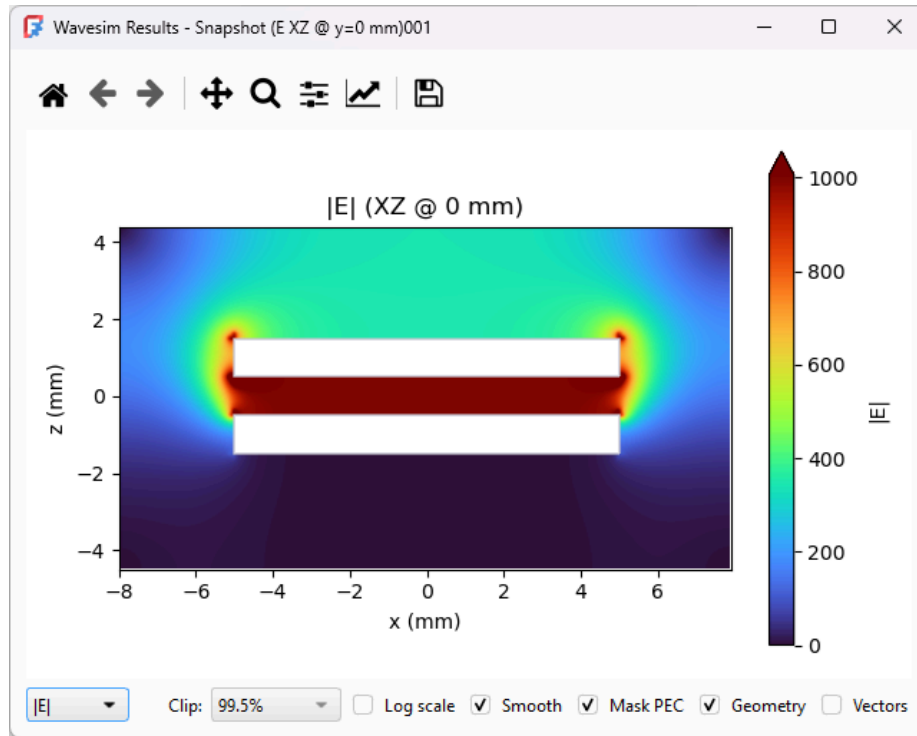


Figure 4.13: The field magnitude $|E|$ on the same plane.

The slices can be checked by eye. In Figure 4.12, ϕ falls linearly from 1 V to 0 V across the gap and wraps around the plate ends. In Figure 4.13 the field in the gap is a flat 1 V / 1 mm = 1000 V/m, and above **Top** about 330 V/m: 1 V over the 3 mm to the grounded lid. Below **Bottom** it is close to zero, because that plate and the floor are both at 0 V. The field peaks at the plate corners, where it is singular in theory and limited in practice by the mesh. The field energy in the window's header, 0.871 pJ, equals $\frac{1}{2}Q_{\text{Top}}V_{\text{Top}}$ to the last digit.

4.2.4 The mesh

Running the same document at three resolutions shows whether 1.024 pF belongs to the mesh or to the geometry:

Table 4.2: Mutual capacitance against resolution, parallel-plate example.

Resolution	Grid	Cells	Mutual C	vs. $\epsilon_0 A/d$
5	$80 \times 80 \times 45$	0.29 M	1.030 pF	+16.4%
10	$160 \times 160 \times 90$	2.3 M	1.024 pF	+15.6%
15	$237 \times 237 \times 130$	7.3 M	1.022 pF	+15.4%

The value falls steadily, and ever more slowly, towards about **1.020 pF**: a 15% fringing correction, which the default resolution reaches to within 0.4%. The closed form is not wrong; it describes a simpler configuration than these plates.

Unlike the coax, this model needs no refinement. The plates lie exactly on grid lines, so the Report view states that conformal PEC is not in effect: no cell is cut, and the staircase is already exact (Section 7.4).

Chapter 9 covers the electrostatic solver, the Set Potential panel and the capacitance window in full.

Command reference

The workbench adds two toolbars and one menu. The main toolbar, **Wavesim**, holds 18 buttons in five groups separated in the order they are used: set up the model, drive it, measure it, run it, configure it (Figure 1.1). The second toolbar, **Wavesim Cross Section**, occupies its own row below the first; it cuts the model open for inspection and changes nothing in it (Section 5.6). The **Wavesim** menu lists the same commands, the two cross-section buttons included.

Four rules hold for every button:

- **Most buttons require a Simulation**, and are greyed out until one exists. Only `New Simulation` and `Settings...` act without one. `Refine Body Mesh` additionally requires a selected solid, and `Settings...` is never greyed out.
- **An Add button creates its object and opens its panel.** The object is placed in the Sources or Monitors group. OK keeps it, **Cancel deletes it again**. Double-clicking it in the tree reopens the panel. The voltage and current monitors are the exception and have no panel.
- **A new object starts at the middle of the domain.** A point source and a probe are created at the centre of the domain box, a snapshot on the XY plane through that centre, and a modal port or Gaussian beam on the Z min face. Each is moved from its own panel.
- **Refusals are printed in the Report view** (View → Panels → Report view), each line beginning `Wavesim: .`

5.1 Model setup

Table 5.1: The Model setup group.

Button and its tooltip	
	New Simulation “Create a Wavesim simulation container in the active document”
	New Material “Create an EM material (eps_r / mu_r / sigma / PEC); drag bodies onto it in the tree to assign them”
	Refine Body Mesh “Mesh the selected body with a chosen number of cells across its bounding box in x/y/z”

`New Simulation` builds the Simulation with its Domain, its groups and the two seeded materials, Vacuum and PEC (Section 3.1). With no document open it creates one first.

`New Material` opens the material panel. **The material is created only on OK**; Cancel leaves nothing behind. Geometry is not selected first: bodies are assigned by **dragging them onto the material in the model tree** (Chapter 7).

`Refine Body Mesh` requires one or more solids to be selected — in the tree, or by clicking any face of the body in the 3-D view, which selects the whole body — and is greyed out until they are. A body that is not assigned to a material is not part of the simulation, and is refused:

These bodies are not assigned to a Material, so they are not part of the simulation and cannot be refined: ... Drag them onto a Material in the tree first.

The panel sets the number of cells across the body’s bounding box (Section 6.6).

5.2 Sources and ports

Table 5.2: The Sources and ports group.

Button and its tooltip	
	Add Point Source “Add a soft point source with a selectable temporal excitation (Gaussian, sine, rectangular, Gaussian+sine)”
	Add Modal Port “Add a modal waveguide port that terminates a domain face with the TEM mode of the PEC cross-section there — launching the mode inward and absorbing what returns (exactly, and at DC), with no PML needed on that face. Driven by a temporal waveform or an external SPICE circuit (chosen in the task panel)”
	Add Gaussian Beam “Add a directional Gaussian-beam source launched from a domain face, with a selectable polarization, waist and temporal excitation”
	Add Lumped Port “Add a lumped R/L/C element (optionally with a voltage or current source) across a gap, between two picked vertices/faces or along a picked edge”
	Add SPICE Line Port “Add a lumped port across a gap, coupled in lockstep to a user ngspice netlist (SPICE co-simulation)”
	Set Potential “Electrostatic mode: hold PEC bodies at potentials. Bodies touching a driven one inherit it; the rest are grounded at 0 V”

The first five buttons drive a full-wave run and the last an electrostatic one, so the group is never available in full: the five are greyed out in electrostatic mode, as is **Port Matrix (Z / Y)**, and **Set Potential** in full-wave mode. The toolbar keeps its shape across a change of mode.



Figure 5.1: The toolbar in electrostatic mode: the sources greyed, Set Potential live.

Add Point Source creates a source at the domain centre. Its panel sets the position, the field component and the waveform (Section 8.2).

Add Modal Port creates a port on the Z min face; the face is chosen in its panel, which also selects between a waveform drive and a SPICE drive. There is no separate SPICE button for a modal port (Section 8.3).

Add Gaussian Beam launches a beam from the Z min face (Section A.1).

Add Lumped Port takes the port’s two terminals from the current selection, so **the gap’s geometry is selected before the button is pressed**: two vertices, two flat faces, or one edge. The terminals can be changed afterwards in the panel (Section 8.4).

Add SPICE Line Port opens its panel. The port’s line is set by **dragging a two-point sketch onto the port in the tree** (Section A.2).

Set Potential opens a panel listing every PEC body. It creates no object: it edits the bodies’ own settings and writes nothing until OK is pressed (Section 9.3). With no PEC body in the document it refuses:

Wavesim: no PEC bodies to hold a potential. Assign solid bodies to a material with PEC ticked.

5.3 Monitors

Table 5.3: The Monitors group.

Button and its tooltip	
	Add Probe “Add a point field probe that records a field value over time”
	Add Snapshot “Add a snapshot monitor capturing a 2D field slice at a chosen time interval”
	Add Energy Monitor “Record the total electromagnetic energy in the domain over time, with or without the PML volume”
	Add Dissipation Monitor “Record the ohmic power absorbed by lossy material over time, with or without the PML volume”
	Add Voltage Monitor “Record the voltage $V(t) = \int \mathbf{E} \cdot d\mathbf{l}$ along an open sketch curve (drag the sketch onto the monitor in the tree)”
	Add Current Monitor “Record the current $I(t) = \oint \mathbf{H} \cdot d\mathbf{l}$ around a closed sketch curve (drag the sketch onto the monitor in the tree)”

A probe is created at the domain centre and a snapshot on the XY plane through it. What each monitor records is described in Section 8.5. The monitor buttons are not greyed out in electrostatic mode, although that solve has no time axis: it ignores point sources, modal ports, lumped ports and probes, and prints a warning naming what it ignored (Chapter 9).

One energy monitor and one dissipation monitor are permitted per document. Both buttons remain enabled, but a second press is refused with `Wavesim: an energy monitor already exists.` or `Wavesim: a dissipation monitor already exists.`

The voltage and current monitors have no panel

Either button creates the monitor in the tree and prints an instruction in the Report view; there is no panel to open. The path is set by **dragging a sketch from the model tree onto the monitor**:

- voltage — an **open** curve. $V = \int \mathbf{E} \cdot d\mathbf{l}$ runs from the curve’s start to its end.
- current — a **closed** curve. $I = \oint \mathbf{H} \cdot d\mathbf{l}$ is positive by the right-hand rule along the curve’s direction.

A monitor with no sketch dropped on it records nothing.

5.4 Solving

Table 5.4: The Solving group.

Button and its tooltip	
	Run Simulation “Run the simulation out-of-process via the Wavesim solver and load the results”
	Port Matrix (Z / Y) “Run the solver once per selected port and extract the Z / Y matrix relating them”

`Run Simulation` performs four steps in order:

1. Where the document already has results on disk, it asks first:
 - `Previous results for this document will be overwritten.` `Overwrite and Run` proceeds;
 - `Cancel` allows the old results to be moved aside.
2. It turns the geometry into a grid, in a progress window with a `Cancel` button.

3. It starts the solver and reports its progress. **FreeCAD does not respond while the solver runs**; only the progress window updates.
4. It rebuilds the Results group and shows a **Wavesim Run Complete** summary.

A document with no body assigned to a material runs a small demo box instead. Section 8.6 describes a run in detail.

Port Matrix (Z / Y) runs the solver **once per selected port**, and therefore takes N times as long as a single run. It requires at least two ports that record their own $V(t)$ and $I(t)$ — modal or lumped ports — and refuses otherwise. A dialog selects the ports, after which the same overwrite question is asked. The result is a **Port Matrix** group beside Results, holding a Z or Y matrix rather than S -parameters (Section A.3).

Four buttons remain enabled and refuse when pressed, each stating its reason in the Report view: a second energy or dissipation monitor, Set Potential with no PEC body, Port Matrix (Z / Y) with fewer than two ports, and Refine Body Mesh on a body assigned to no material.

5.5 Settings

Table 5.5: The Settings group.

Button and its tooltip	
	Settings... “Configure the Wavesim solver interpreter and repository paths”

Opens the Settings dialog (Section 2.3). It requires no document and is never greyed out.

5.6 The cross-section toolbar

The cross-section toolbar cuts the model open on a plane and draws the cell grid on the cut, which is how the mesh is inspected against the geometry it is meant to resolve. It carries two on/off buttons and three controls:

Table 5.6: The cross-section toolbar’s controls.

Control	What it does
	Cross Section On/Off Cuts the model on the plane, or restores it.
	Mesh Grid On/Off Draws the Domain’s cell grid on the cut plane.
Axis (X / Y / Z)	Which axis the cut plane is normal to.
Position (mm)	Where the cut plane sits on that axis.
Flip	Which half of the model is kept; the cut plane does not move.

The two buttons are linked: the grid is drawn only on a cut, so switching the grid on switches the cut on with it, and switching the cut off takes the grid down.

The cut plane is the domain’s grid plane. There is one position, not two, and moving the cut moves the drawn mesh with it. While the grid is on, only the grid plane on the cut axis is drawn; switching the cut off hides all three. The first cut in a document is normal to Z, through the middle of the domain. Axis, position and Flip can be set before the cut is switched on, and are stored with the document.

The cut itself. Each visible solid is cut into a closed solid, with a capped face where the plane passed, so a cut conductor reads as metal rather than as a hollow shell. The status bar reports how many bodies were cut. A body lying entirely on the discarded side is hidden, cut or not, and where the plane lies beyond every body nothing remains to draw, as the Report view states.

Its limits. The cut is a view: a cut model runs exactly as an uncut one, and the cut cannot enlarge the domain. It is not live, so geometry or mesh edited while a cut is up appear when the cut is switched off and on again. It does not survive closing the document, which reopens whole with the cut off.

Axis, position and Flip are not commands: they have no menu entry, no keyboard shortcut, and FreeCAD's Customize dialog does not list them. The **cap style** has no toolbar control at all. It is the Domain property `WavesimXSecStyle`, in the property editor's Grid group: `Shaded` (the default) tints the cut face a darker shade of the body's colour, `Hatched` draws cross-hatching over it, and `Plain` leaves it in the body's colour.

The domain and the mesh

The **Domain** defines the region that is solved: the extent of the box, the condition on each of its six faces, and the grid of cells inside it. It is created with the Simulation and listed first beneath it, and double-clicking it opens its panel. Edits take effect immediately: the cell counts and the grid in the 3-D view follow each change. Cancel restores the state the panel opened with.

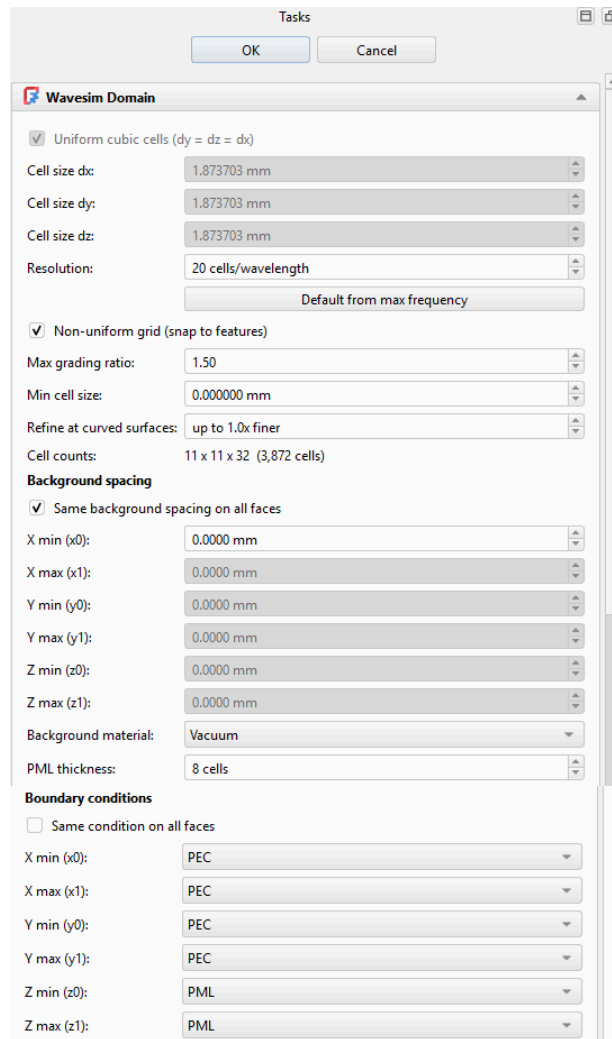


Figure 6.1: The Domain panel, full wave.

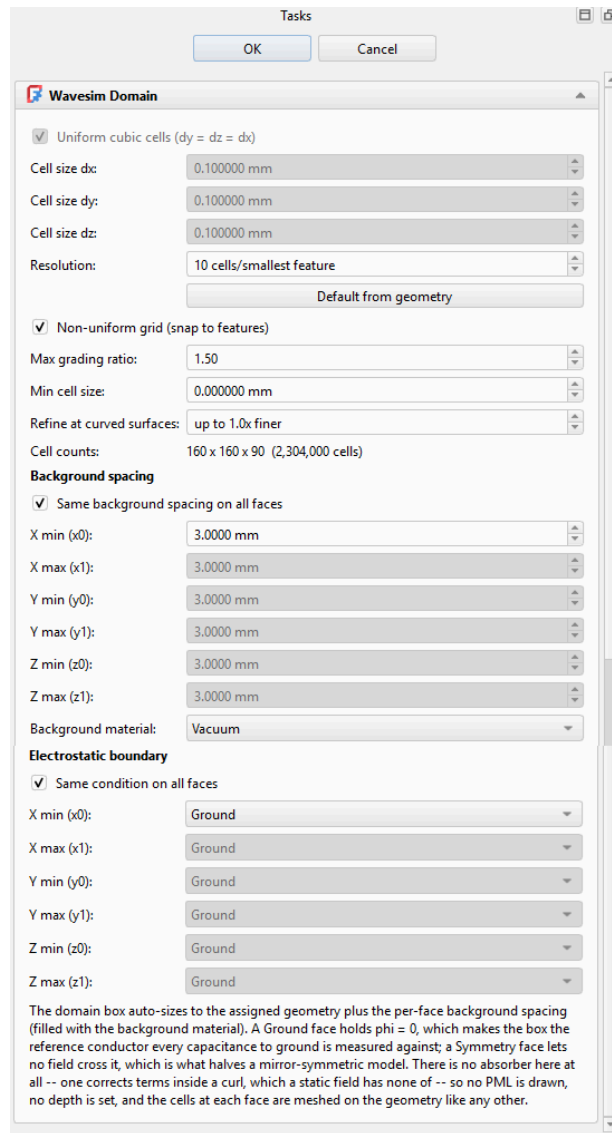


Figure 6.2: The Domain panel, electrostatic.

6.1 The domain box and background spacing

The box is the smallest box that contains the contents of the simulation (Section 3.3), extended on each face by a **background spacing**. The panel has one spacing row per face under **Background spacing**, each default **0 mm**, and **Same background spacing on all faces** drives all six from the first. The spacing is filled with the **background material** selected in the same panel (Chapter 7).

The 3-D view draws the box as a blue wireframe. It encloses the entire grid, and nothing outside it is solved. An orange box inside it marks the inner edge of the absorber (Section 6.3). The Domain's eye icon hides both boxes and leaves the mesh drawn, so the mesh can be compared with the geometry alone.

Unlike the other panels, the Domain panel does not grey out a row bound to an expression (Section 3.4). Edit such a property through its expression: a value typed into the panel is replaced at the next recompute.

6.2 Boundary conditions

Each face has its own condition, chosen from the set the solver mode offers:

Table 6.1: The boundary conditions offered in each solver mode.

Mode	Choices	Default	Meaning
Full wave	PML , PEC	PML	PML absorbs outgoing waves. PEC is a perfectly conducting wall.
Electrostatic	Ground , Symmetry	Ground	Ground holds $\phi = 0$ and makes the box the reference conductor. Symmetry lets no field line cross the face: a mirror plane.

Both sets are stored, so switching mode loses neither. `PML thickness:` sets the absorber depth, default **8 cells** electrostatic mode has no absorber and hides the row. `Same condition on all faces` drives every face from the first.

Two further conditions are set by an object on the face rather than chosen, and appear as greyed-out entries:

- **Modal port** — a face that carries a modal port. The port itself terminates the face, so the face has no PML, no PEC wall and no background spacing; its spacing row is taken as zero whatever it holds. Deleting the port returns the face to the ordinary conditions.
- **Port / beam (absorbing)** — a face from which a Gaussian beam or a SPICE-driven modal port launches (Appendix A). The source launches from a plane just inside the face, so the face behind it is always absorbing. Deleting the source returns the face to the ordinary conditions.

6.3 Position of the absorber

The absorber is inside the box, not around it

The PML occupies the outermost cells of the drawn box, **8** of them by default; it is not a shell added outside it. Three consequences follow.

- The background spacing is the only clearance between the geometry and the absorber, and it is 0 mm until set. On a PML face without spacing, the outer surface of the model lies inside the absorber. Such a face needs spacing greater than the absorber depth, plus a margin in which the near field of the structure can decay.
- Geometry carried straight out through a face at constant cross-section is absorbed as though it continued to infinity. This is what zero spacing is for: it is how a line, substrate or trace is made infinitely long.
- A body that ends, bends or changes cross-section inside the absorber reflects from its own discontinuity. The result looks like a poor absorber, not like a modelling error.

The workbench checks for this twice. The Domain panel shows a live red notice beginning In the PML: , judged from the bodies' bounding boxes:

In the PML: Trace ends inside the x1 absorber (42 to 50 mm). A PML terminates what runs straight out through it; a body that stops or changes section in there reflects. Carry it out to the face, or raise that face's background spacing past 8 mm.

At run time the Report view repeats the check on the voxelised cells, which also catches a cross-section that varies inside the absorber:

Wavesim: the x1 absorber is not invariant along x: ... cells in its 8-cell shell differ from the section at its inner edge ...

6.4 What sets the cell size

In full wave the cell size follows from the wavelength:

$$\Delta = \frac{c}{f_{\max} \cdot N_{\lambda} \cdot \sqrt{\epsilon_r \mu_r}},$$

where $f_{\max}$ is the Simulation's maximum frequency and N_λ the Resolution: row, in cells per wavelength (default **20**). With the graded mesh (Section 6.5), ϵ_r and μ_r are those of the background material, and each dielectric body is meshed at its own, shorter wavelength. The grid is rectilinear, so that finer spacing runs across the whole slab the body spans on each axis, not across the body alone. With a uniform grid, ϵ_r and μ_r are those of the highest-index non-PEC material, so the one cell size is fine enough everywhere.

An electrostatic run has no frequency, and therefore no wavelength. The field varies fastest across the thinnest part of the model, so the cell size is the thinnest bounding-box dimension of any material body, conductors included, divided by the Resolution: row, here in cells per smallest feature (default **10**). A body with no thickness along an axis, such as a flat sheet, is judged by its other two dimensions.

Both resolutions are stored, so switching mode loses neither. The cell size is derived again whenever one of its inputs changes: the resolution, the background material or the grid mode in the Domain panel; the maximum frequency or the solver mode, when the Simulation panel is accepted; and, in electrostatic mode, the set of bodies assigned to materials. The Default from max frequency button (Default from geometry in electrostatic mode) repeats the same derivation on demand; it is never required.

With the graded mesh, the Cell size dx/dy/dz: rows are greyed out and show the derived background size. With a uniform grid a size can be typed in, and Uniform cubic cells (dy = dz = dx) keeps the three equal. A typed size is replaced whenever one of the inputs above changes.

6.5 The graded mesh and its controls

Non-uniform grid (snap to features) is **on** in a new Domain. Grid lines are then placed on the geometry itself, so no cell is half metal:

- on every face of each body's bounding box;
- on flat faces normal to an axis: the walls of a slot, pocket or step;
- on the silhouettes of cylinders parallel to an axis;
- on the outlines where two faces meet, such as the circles bounding a bevel;
- through the centre of a full circle, where this adds no cells.

The intervals between these lines are filled with cells no larger than the target size, graded up to the background size away from fine features. Absorber cells stay uniform at the background size: nothing is snapped or refined inside the absorber. Three rows set the balance between accuracy and cost:

Table 6.2: The graded-mesh controls.

Row	Default	What it does
Max grading ratio:	1.5	The largest size ratio between neighbouring cells. A larger ratio grades faster and uses fewer cells.
Min cell size:	0 mm (no limit)	The smallest cell the mesher may make. Feature lines closer together than this are merged, which moves an interface by up to half this size. It protects the time step (Section 6.8).
Refine at curved surfaces:	up to 1.0x finer (off)	A gently curved surface tangent to a grid plane — a sphere’s pole, a cylinder’s side — otherwise collapses into a single flat step. Values above 1 allow cells that much finer there. The refinement adds whole planes of cells, so modest values are best.

6.6 Refining one body

Where the automatic mesh is too coarse across a body — usually the cross-section of a transmission line — **Refine Body Mesh** sets the cell count across the selected body directly. The panel takes **Cells across X:**, **Y:** and **Z:**, counted across the body’s bounding box, and opens with the counts the body already holds. A count of 0 reads **off** and makes no request on that axis. As the counts are typed, the panel reports the resulting target **Cell size:** and the whole **Grid:** — cell counts, finest cell and time step **dt**. **Clear refinement** returns all three to 0.

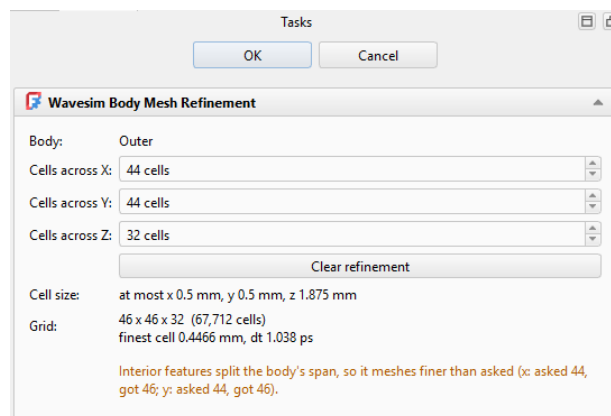


Figure 6.3: The Refine Body Mesh panel, with its grid readout.

The request is stored on the body, as the properties **WavesimCellsX**, **Y** and **Z**. It moves with the body, survives reassignment to another material, and can be bound to an expression for a sweep. Bodies selected together receive the same request. OK is a single undo step; Cancel restores the previous requests.

The count is a lower bound. Interior features such as holes or slots divide a body with their own grid lines, and each part receives at least its share, so the body can end up with more cells than requested. A request can make a body’s cells finer, never coarser. The panel warns in orange when the request cannot be met as stated: when **Min cell size** is coarser than the requested cells, when the graded mesh is off (the request is then ignored), and when interior features give more cells than asked.

6.7 Looking at the mesh

The Domain draws its grid on three planes, one normal to each axis, placed by the properties `GridPlaneX`, `Y` and `Z` and switched on by `ShowGridPlaneX`, `Y` and `Z`, all **off**. The cross-section toolbar (Section 5.6) drives these properties: it cuts the model open on one plane and draws that plane's grid, so the cells are seen against the true cross-section.

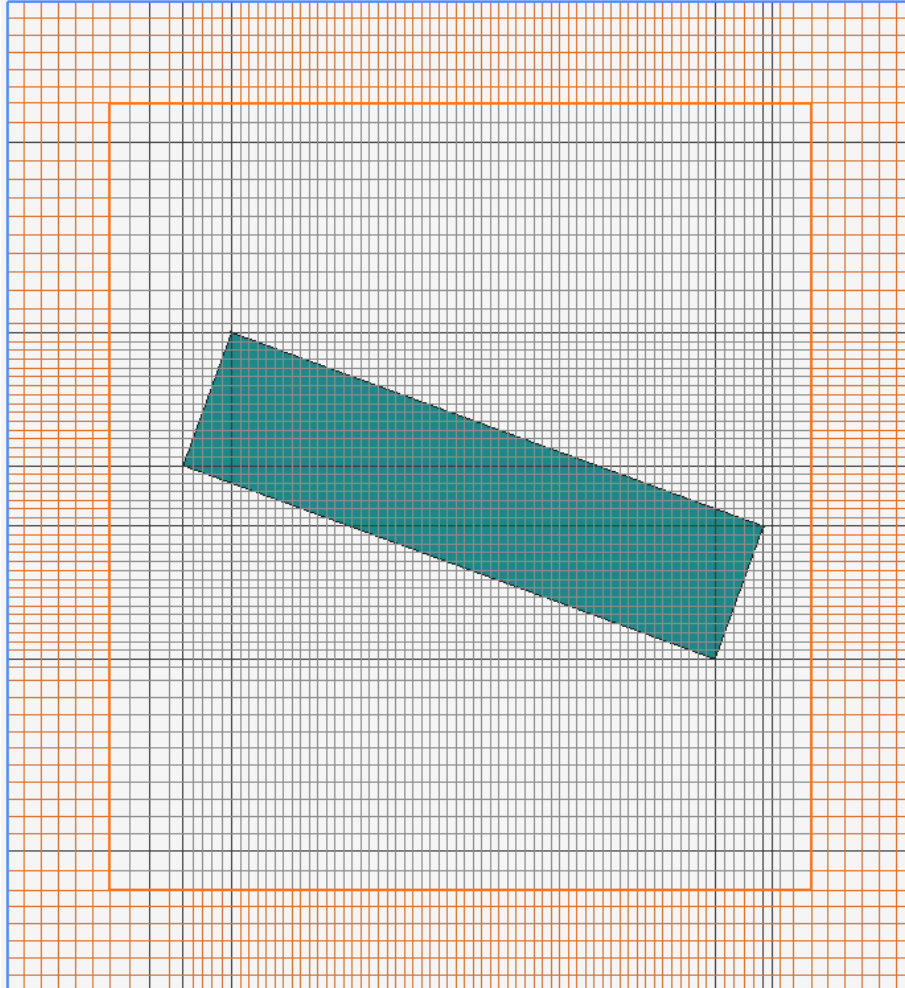


Figure 6.4: A grid plane through a tilted body. The blue box is the domain, the orange box the inner edge of the absorber, and the orange lines the absorber's own cells, which are uniform at the background size. The graded cells lie on the body's bounding box, not on its inclined faces.

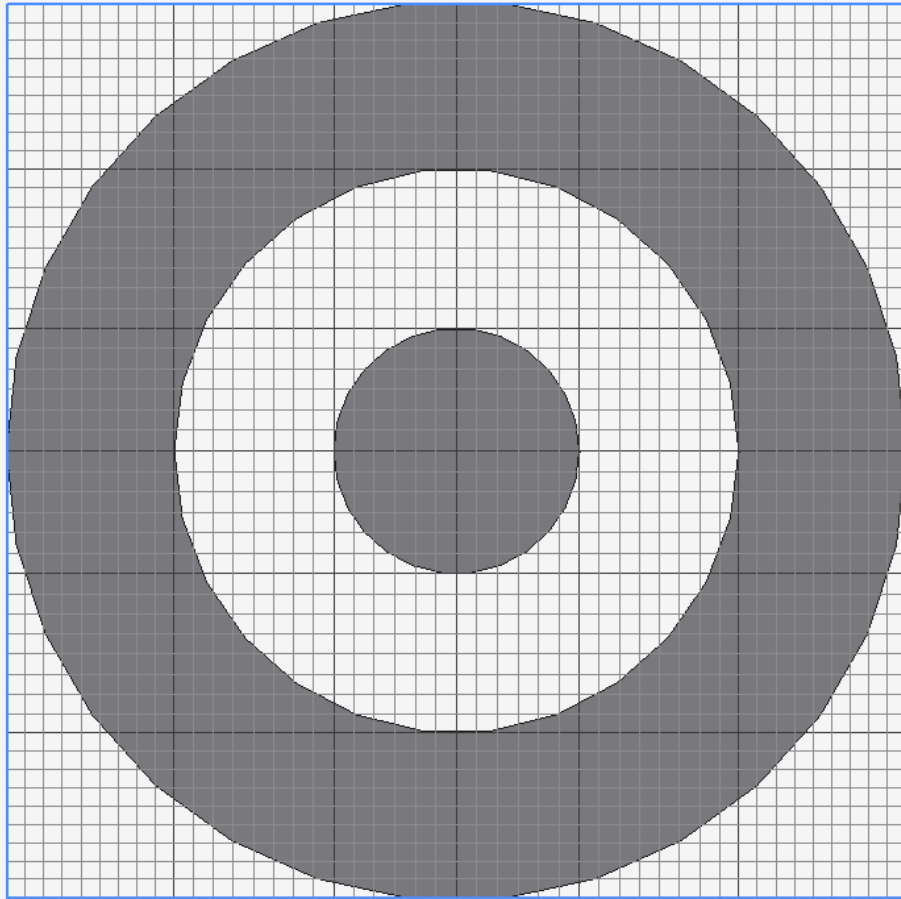


Figure 6.5: A cross-section of the coax, with the mesh drawn on the cut.

The line colour shows where each line comes from: **dark** lines are placed on geometry, **grey** lines fill the intervals between them, and **orange** lines lie in the absorber. Past 400 lines along an axis, the drawing shows only every few lines, so a dense grid can look coarser than it is. The lines placed on geometry have an allowance of their own, so the thinning never drops one of them. Only the drawing is thinned, never the mesh.

6.8 The cost of a fine cell

A full-wave run advances in fixed time steps Δt , and the grid limits the step (Section 8.1):

$$\Delta t = \frac{0.99}{c \sqrt{\frac{1}{(\Delta x_{\min})^2} + \frac{1}{(\Delta y_{\min})^2} + \frac{1}{(\Delta z_{\min})^2}}},$$

where each $\Delta_{\min}$ is the narrowest cell anywhere along that axis. A single fine cell therefore sets the step for the whole domain. With 1 mm cubic cells, $\Delta t = 1.91$ ps. A single layer of 0.1 mm cells anywhere in the domain reduces it to 0.33 ps, so the same simulated time takes 5.8 times as many steps, in every cell of the model. Run time scales as the number of cells times the number of steps.

This is why the graded mesh refines only where the geometry needs it, why curvature refinement is off by default, and why Min cell size exists. The Simulation panel's Time steps: row and the Refine panel's dt both report the cost before a run. An electrostatic solve has no time step, so there a fine cell costs only its share of the cell count.

The graded mesh is capped at 10 million cells. A graded mesh over the cap is coarsened as a whole, step by step, until it fits. No message is printed; the cell count in the Domain panel is the only sign. A uniform grid over the cap stops the run instead, with this error in the Report view:

```
Voxel grid too large: ... cells (limit 10,000,000). Use a coarser grid or smaller geometry.
```

Materials

A **material** is a set of electromagnetic constants together with the bodies made of it. A body not assigned to a material takes no part in the simulation (Section 3.3).

7.1 Creating a material and assigning bodies

New Material opens the material panel, which takes a name, a colour and the constants. The material is created only on OK; Cancel leaves nothing behind. Every new Simulation starts with two, **Vacuum** and **PEC** (Section 3.1).

A body is assigned by dragging it onto the material in the model tree. No selection is needed beforehand, and only solids are accepted. An assigned body is listed beneath the material, as in Figure 4.3, and takes its colour. A body belongs to one material at a time: dropping it onto a second material moves it there. Dragging it out of the material and back into the document leaves it unassigned and grey. The material's eye icon shows and hides all its bodies together.

Where bodies of different materials overlap, the shared cells take the material of whichever body is later in the tree. That rule is not guaranteed; a Part boolean that cuts one body out of the other gives each cell a single owner.

7.2 Permittivity, permeability, conductivity, PEC

Table 7.1: The material properties, their panel rows and defaults.

Property	Panel row	Default	Notes
Eps	Relative permittivity (eps_r):	1.0	The panel accepts 1 and above.
Mu	Relative permeability (mu_r):	1.0	
Sigma	Conductivity (S/m):	0.0	Makes a lossy dielectric (Section 7.3).
Pec	Perfect electric conductor (PEC)	off	Greys out the three rows above.

The constants are frequency-independent; there is no dispersion model. A PEC body holds $E = 0$ inside it. In a full-wave run it acts as a boundary condition; in an electrostatic run it is a conductor held at a potential. That potential is a property of the **body**, not of the material, because one PEC material commonly holds both a trace and its ground plane, which must be at different potentials. Potentials are set with Set Potential (Section 9.3), which the Set potentials... button on a PEC material's panel also opens.

μ_r and modal ports. The mode solve behind a modal port (Section 8.3) assumes $\mu_r = 1$ on the port plane, so a magnetic material there leaves the reported Z_0 , L and v unchanged. The full-wave run itself does use μ_r .

7.3 Lossy dielectrics and the accuracy limit

A non-zero conductivity gives a lossy dielectric: silicon, soil, tissue, sea water. A good conductor, however, is modelled as PEC, not as a large conductivity, because of how the time-stepping treats loss. Each step damps E in a lossy cell by the factor $(1 - k)/(1 + k)$, with

$$k = \frac{\sigma \Delta t}{2\epsilon_0 \epsilon_r},$$

that is, half a time step measured in units of the relaxation time $\varepsilon_0\varepsilon_r/\sigma$. Above $k = 1$ the factor is negative: the field flips sign at every step instead of decaying, and dies away only very slowly. The run completes and looks normal, but is wrong. The conductivity of a metal lies far inside that regime. With 1 mm cells ($\Delta t = 1.91$ ps, Section 6.8) the limit is $\sigma \approx 9$ S/m at $\varepsilon_r = 1$ and about 740 S/m at $\varepsilon_r = 80$, while copper, at 5.8×10^7 S/m, gives $k \approx 6 \times 10^6$.

The material panel computes k on the current grid as the constants are typed. Below the limit it shows

```
Lossy dielectric: sigma*dt/(2*eps0*eps_r) = ... (must stay under 1). Conductivity does not affect the timestep.
```

Above it, the line turns red and begins

```
Material '...': sigma=... S/m at eps_r=... gives sigma*dt/(2*eps0*eps_r) = ... > 1 on this grid's timestep.
```

The same sentence appears in the Report view at run time, but the run is not stopped. A finer grid lowers Δt , and with it k ; for a real metal, the remedy is PEC.

Loss does not shorten the time step. A lossy material restricts the run to the CPU (Section 11.3).

7.4 Conformal PEC and subpixel smoothing

A body's surface rarely coincides with cell faces. Without correction, each cell is assigned wholly to one material or the other, which renders the surface as a staircase. Two options on the Simulation panel reduce the resulting error:

- Conformal (cut-cell) PEC conductors, **on** in a new simulation. A cell face partly covered by a conductor is integrated over its open area, rather than counted as wholly metal or wholly open. This removes the first-order error in a line's impedance, and the spurious higher-order mode that a staircased conductor launches at a modal port. In return, voxelisation takes longer, because conductor surfaces are sampled finely. Cells that a conductor barely clips are clamped for stability, at a threshold the run summary reports.
- Subpixel smoothing of dielectric interfaces, **off** in a new simulation. A cell cut by a dielectric boundary receives an averaged, direction-dependent permittivity, so results vary smoothly as the geometry moves. PEC is unaffected.

Conformal PEC also restricts the run to the CPU (Section 11.3).

7.5 The background material

Every cell that no body fills, including the background spacing (Section 6.1), holds the Domain's **background material**, selected in the Domain panel. A new Simulation uses **Vacuum**. Any material can be selected. Where the document has no vacuum material ($\varepsilon_r = \mu_r = 1$, not PEC), the list also offers Vacuum (eps=1, mu=1). The background sets the coarse cell size of the graded mesh (Section 6.4).

A PEC background fills the box with metal, which dielectric bodies then displace where they sit: one way of modelling the interior of a waveguide or cavity. Conformal PEC does not apply to a metal background, whose walls follow the cells as a staircase.

Full-wave (FDTD) simulations

A full-wave run steps Maxwell's equations through time on the grid built in Chapter 6. Energy enters through a **source** or a **port**, **monitors** record what follows, and `Run Simulation` starts the run. This chapter covers each of those objects, the quantities its panel takes, and the few rules that, when broken, give a plausible wrong answer rather than an error.

8.1 What the solver is doing

THE PHYSICS • skippable

The Yee grid, leapfrog stepping and the CFL limit

An FDTD solver works from Maxwell's two curl equations alone:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}, \quad \nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E}$$

The Yee grid stores each field component where its curl is easiest to take. The three components of $\mathbf{E}$ sit on the midpoints of the cell edges; the three of $\mathbf{H}$ sit on the centres of the cell faces, half a cell away in every direction. Each face is then ringed by exactly the four $\mathbf{E}$ edges whose circulation gives the curl through it, and vice versa. Nothing is interpolated to take a derivative: the staggering is itself the discretisation.

The same staggering is applied in time. $\mathbf{E}$ is stored at whole steps and $\mathbf{H}$ at half steps, so each update is a centred difference of the other field:

- advance $\mathbf{H}$ half a step, from the curl of the $\mathbf{E}$ around each face;
- advance $\mathbf{E}$ a whole step, from the curl of the new $\mathbf{H}$ around each edge.

The time loop is nothing more than this. No matrix is assembled and nothing is inverted, so the cost of a run is cells $\times$ steps, while memory scales with the cells alone.

The grid limits the step. A disturbance moves at most one cell per step, so the step is bounded by the Courant–Friedrichs–Lewy condition,

$$\Delta t \leq \frac{1}{c \sqrt{(1/\Delta x)^2 + (1/\Delta y)^2 + (1/\Delta z)^2}}$$

which follows the smallest cell in the grid, wherever it lies: one fine cell in a corner sets the step for every cell. Section 6.8 works through the numbers.

Numerical dispersion is a consequence of the discretisation: on a grid a wave travels slightly slower than c , by an amount that depends on its frequency and direction. The error falls as the square of the cell size, which is where the customary 10 to 20 cells per wavelength comes from. At the default 20, the phase velocity is off by a few tenths of a percent.

A short pulse contains every frequency up to roughly its own bandwidth, so the Fourier transform of a single transient run gives the response at every frequency at once. Each impedance curve of Chapter 10 is therefore one run, not one run per frequency.

The absorber, a convolutional PML, is a lossy layer that attenuates what enters it without reflecting at its front face. It absorbs a wave leaving the domain; it does not cleanly absorb a structure that stops inside it, an evanescent field that has not yet decayed, or a mode it was not

designed for. A PEC face is the opposite and far simpler: it forces tangential $\mathbf{E}$ to zero, making a perfect mirror.

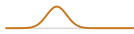
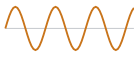
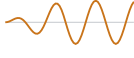
A snapshot frame is not the raw staggered field. Each frame is interpolated to cell centres and to a common instant, so its three components are simultaneous and co-located, and $\mathbf{S} = \mathbf{E} \times \mathbf{H}$, formed from E and H frames of the same instant, is a true Poynting vector.

8.2 Sources and excitation waveforms

A **point source** drives one field component at one point. Its panel has four rows: **Field component:**, one of E_x ... H_z (default $\mathbf{Ez}$), and **Position X/Y/Z:**, a point in world coordinates that is snapped to the nearest grid cell at run time. A point source suits illuminating a resonator or probing a structure's response. It does not suit driving a transmission line: that calls for a modal port (Section 8.3), which launches a known mode at a known impedance.

Below those rows is the waveform, which every drivable object shares. **Excitation:** selects a family, and the parameter rows beneath it change to match:

Table 8.1: The excitation waveform families.

Family	Shape	Parameters	What it is for
Gaussian Pulse		Max frequency, Amplitude	Broadband. The default, and the usual choice for a port: one run covers the band.
Sine Wave		Frequency, Amplitude, Phase offset	Steady state at one frequency. Switches on abruptly, so it rings.
Sinusoid (ramped CW)		Frequency, Amplitude, Phase offset, Ramp-up cycles	The same, eased in over 3 cycles so the turn-on does not ring.
Rectangular Pulse		Amplitude, Start / Rise / Flat-top / Fall time	A shaped gate: step responses, time-domain reflectometry.
Gaussian + Sine		Carrier frequency, Envelope bandwidth, Amplitude, Phase	A narrow band around a carrier: one region of the spectrum at low cost.

The shapes are indicative: they are drawn from each family's formula, not to scale and not by the solver. **Plot excitation vs. time...** draws the actual waveform, with its own parameters, before any run.

The max-frequency row sets the pulse width, since a Gaussian's bandwidth and duration are reciprocal: a higher maximum frequency gives a shorter pulse that carries energy further up the spectrum. Any part of the band above what the mesh resolves is under-resolved, and nothing in the run reports it.

Max frequency on a waveform is a property of its own and does not follow the Simulation's. Meshing for 8 GHz while the pulse stays at a higher value puts most of the injected energy above the band the grid can carry, so set the two together.

Waveform parameters are stored in seconds and hertz, whatever the display units. An expression on one of them is therefore written in SI: `8e9`, not `8` (Section 3.2).

8.3 Modal ports

A **modal port** covers a whole domain face and terminates it with the TEM mode of the conductor cross-section on that face. It launches that mode inward and absorbs what comes back, exactly and down to DC. For the mode it solved, it is a matched load, not an approximation of one. Because the port itself terminates the face, the face has no absorber, no PEC wall and no background spacing (Section 6.2).

A modal port and a PML both terminate a face, but they are different kinds of object.

At each step the port sets the magnetic field just outside its face to the value that a matched continuation of the line would carry. That value is built from the **modal projection** of the field on the plane — the overlap of E with the solved mode profile — so the port terminates the mode and nothing else. For that mode the termination is exact and frequency-independent, down to and including DC, which is where a PML is at its worst. Field on the face that does not project onto the mode is not absorbed at all. On a closed cross-section well below the first higher-order cut-off there is nothing else to absorb, so the difference does not matter. It matters on a face that also carries radiation or an evanescent field; such a face is better terminated by a PML behind the structure.

The two also differ in what the face records. A PML is a passive absorber: it reports nothing and cannot be driven. A modal port carries the mode's field distribution, its characteristic impedance and its ϵ_{eff} , and integrates the modal voltage and current on its plane at every step. Those two series give $Z(f)$ (Section 10.3) and the entries of a port-matrix sweep (Section A.3). When the port is driven, the same plane also launches the mode inward. A PML therefore suits a face that a wave only has to leave, and a modal port a face at which a line is driven or measured.

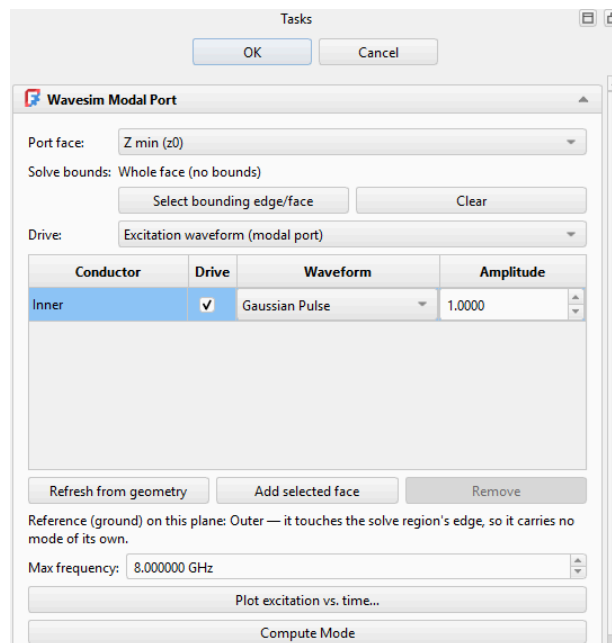


Figure 8.1: The Modal Port panel. **Drive:** selects a waveform or an external SPICE circuit; the table lists one row per conductor on the plane.

The line must reach the port face, and end there

Breaking either rule gives a plausible wrong answer, not an error.

- **The geometry must reach the port face.** With no background spacing, the domain face lies on the end of the line, so the port plane cuts the real cross-section. A line that stops short leaves the port solving the mode of empty space.
- **The conductors must end at the port, not pass through it.** A line carried past its own port puts a second characteristic impedance in parallel with the first, and the run rings for many round trips instead of settling.

8.3.1 Computing the mode

Compute Mode solves the cross-section without an FDTD run. It is a 2-D electrostatic solve on the run's own grid. It needs no time stepping and returns far sooner than a full run, which makes it the quickest way to choose a mesh (Section 4.1.3). For the energised conductor it reports:

Table 8.2: The quantities reported by `Compute Mode` .

Quantity	Meaning
Z_0	Characteristic impedance, $\sqrt{L/C}$.
ϵ_{eff}	Effective permittivity of the mode; 1.000 for an air line.
C, L	Capacitance and inductance per unit length.
v	Phase velocity, also reported as a percentage of c .
f_{max}	The waveform's max frequency, repeated so the plot is self-contained.

C is the only quantity measured. The solver derives $L = 1/(c_0^2 C_{\text{air}})$ from the capacitance of the same cross-section in air, and $Z_0 = \sqrt{L/C}$ from the two. Z_0 , ϵ_{eff} , L and v are therefore one result in several forms. One consequence is that the mode solver ignores μ_r : a magnetic material leaves the derived L unchanged. This is a known limitation of the solver, not of the physics (Section 7.2).

The mode window plots φ over the cross-section with E arrows on top, in model coordinates: the axes are those of the 3-D view. A full run also saves the mode shapes, and they can be reopened from the Results tree (Section 10.5).

8.3.2 The conductor table

The table lists one row per conductor region the port plane cuts, named after the body it belongs to. Each row has its own `Drive` tick, waveform and amplitude, and the port launches the sum of its driven rows. Each row also records its own $V(t)$ and $I(t)$, which is what lets each conductor of a multi-conductor port enter a port matrix.

One conductor never receives a row: the **reference**, which the panel names beneath the table. A conductor that touches the edge of the solve region carries no mode of its own. It is the reference the others are measured against, because a mode is a potential difference between conductors.

`Refresh from geometry` rebuilds the table after the geometry changes. `Add selected face` adds a conductor the automatic scan missed; select its face in the 3-D view first. `Remove` deletes a row.

An undriven row is not an absent row

Unticking `Drive` sets a row's amplitude to 0: the row launches nothing but still terminates its mode. Deleting the row instead leaves that mode with no absorber, on a face that has no PML, and the mode reflects off the boundary.

A passive, receive-only port is therefore a valid object, and it is how the far end of a line is terminated: add the port and leave its drive unticked. It still records $V(t)$ and $I(t)$.

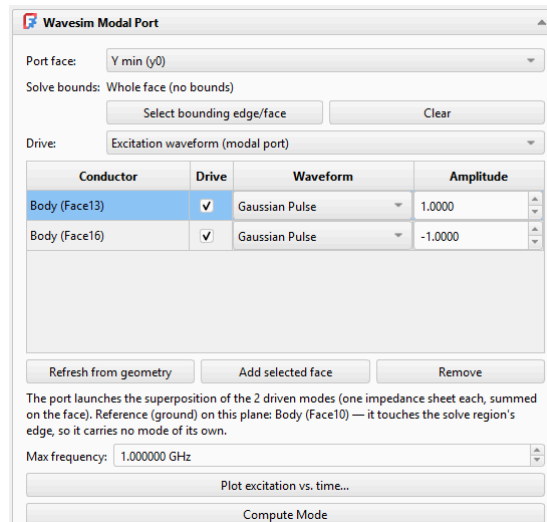


Figure 8.2: A two-conductor port driven differentially: both rows ticked, at amplitudes +1 and −1. The reference conductor is named beneath the table.

Solve bounds: confines the mode solve to a rectangle within the face, taken from the bounding box of a selected edge or face; **Clear** restores **Whole face (no bounds)**. It is for a face that carries two structures which should not be solved as one system.

8.3.3 What a port records

The port records $V(t)$ and $I(t)$ for each row. V is the modal projection of $\mathbf{E}$ on the plane, and I the projection of $\mathbf{H}$ one cell inside it, paired with V so that VI is the power the mode carries. Both are taken at the same instant and counted positive into the domain. They are stored with a **reference impedance** Z_{ref} , which turns them into a wave pair: the incident and outgoing amplitudes are $a = (V + Z_{\text{ref}}I)/2$ and $b = (V - Z_{\text{ref}}I)/2$. Z_{ref} is the mode's own Z_0 , and the impedance plots mark it (Section 10.3).

8.4 Lumped ports

A **lumped port** is an R/L/C element on a line across a gap, with an optional ideal source. It adds a termination, a feed or a discrete component to a structure without modelling the component's physical form.

The geometry comes from the selection. The **+** and **-** terminals are set from two vertices, two parallel flat faces or an edge, and the element is the straight line between them. Positive port current leaves at the **+** terminal; **Swap + / -** exchanges the two.

The three branches are independent, each with a tick and a value:

Table 8.3: The three branches of a lumped port.

Branch	Default	State	Unticked means
Resistance	50 Ω	on	a short (series) or an open (parallel)
Inductance	1 nH	off	the same
Capacitance	1 pF	off	the same

Branches wired: is **Series** or **Parallel**, and has no effect with a single branch. **Drive:** defaults to **None (passive load)** the alternatives are **Voltage source** (Thévenin, in series with the load) and **Current source** (Norton, in parallel with it), each taking a waveform from Section 8.2.

An unticked branch is absent, not zero

In series, an unticked branch is a **short**; in parallel, an **open**. Unticking the resistance of a parallel R–C port leaves the capacitor alone; it does not put $0\ \Omega$ across the capacitor, which would short it. The tick says whether the component is present, not what its value is.

Two placement rules decide which Yee edges the line covers:

- **The line stays out of the metal.** Select the faces that bound the gap — a trace’s underside, a ground plane’s top — not the far side of either conductor. On an edge inside a conductor, the metal wipes the element’s charge at every step. This reads back as a large spurious series capacitance, and nothing looks wrong.
- **The line reaches both conductors.** An element that stops short of the metal puts the remaining sliver of gap in series with it: a few femtofarads, which is kilohms at 10 GHz.

Beyond these two rules the line needs no adjustment: each Yee edge is weighted by the length of line inside its own cell, so an uneven mesh is handled correctly.

The element delivers exactly the value typed, with nothing to subtract or compensate. What does lie across it is the gap’s own capacitance: the cells the line bridges hold $C_{\text{cell}} = \epsilon dA/dl$ in parallel, the real capacitance of a real gap of that shape, present with or without the element. The panel reports it as a fraction of the load. It is an artefact only where the component’s footprint is smaller than one cell; the remedy is then a finer mesh across the line, never a change to the value.

The panel’s **Geometry:** row is a live report: the endpoints it found, the length, whether the line runs along a grid axis, and C_{cell} as a fraction of the load. It warns when that fraction is large.

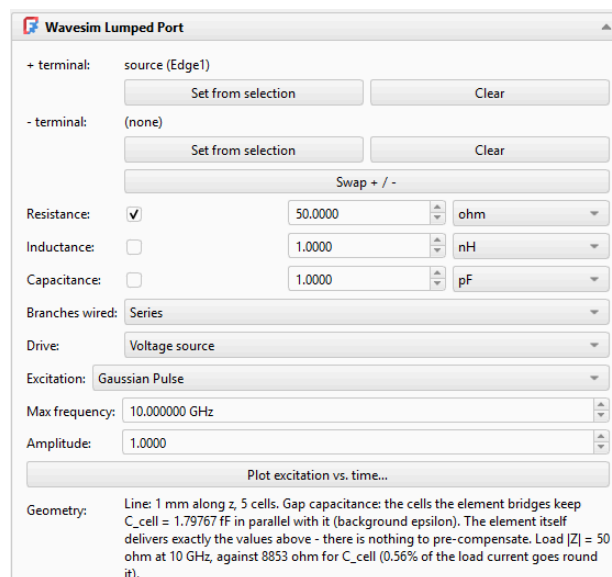


Figure 8.3: The Lumped Port panel, with its live geometry report.

8.5 Monitors

Monitors decide what the run records. Apart from what the ports record themselves, nothing is saved that no monitor asked for.

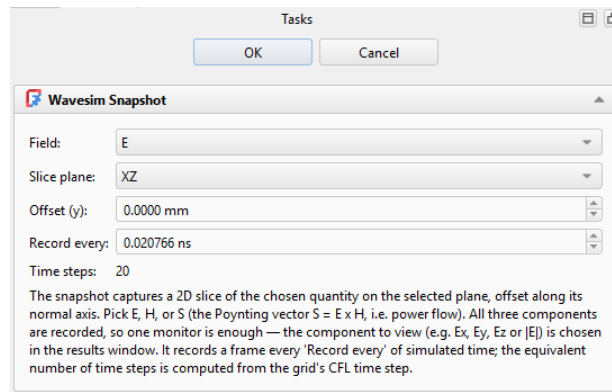


Figure 8.4: The Snapshot panel.

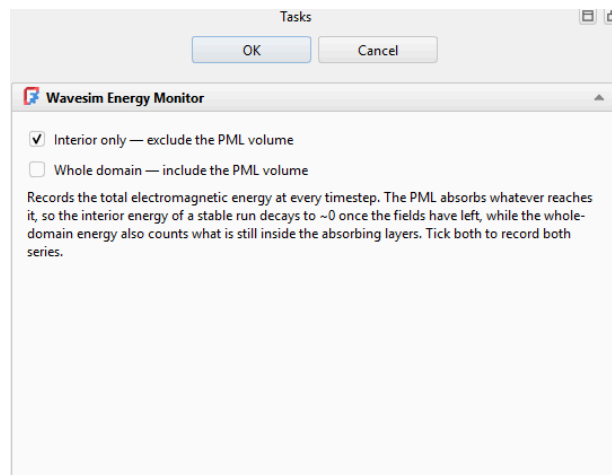


Figure 8.5: The Energy panel.

Probe — one field component, or a magnitude, at one point. **Field quantity:** offers $E_x \dots H_z$ plus $|E|$ and $|H|$ (default E_z), and the three position rows give a point in world coordinates. It is the cheapest monitor.

Snapshot — a 2-D slice, recorded as a movie. The choice is a **field**, not a component: **Field:** is E , H or S , the Poynting vector $\mathbf{E} \times \mathbf{H}$. All three components are recorded, and the component to display is chosen in the results window, so one monitor per field is enough. **Slice plane:** is XY , YZ or XZ , with an offset along its normal. **Record every:** is a simulated time, not a step count. The panel converts it and shows the equivalent step and frame counts beneath, which is the amount of data the run will write.

Energy and **Dissipation** have no position: they sum over a volume. Two checkboxes choose which:

- **Interior only** — exclude the PML volume, default **on**
- **Whole domain** — include the PML volume, default off.

Both can be ticked, and both series are then recorded. Unticking the only ticked box ticks the other, since a monitor with neither would record nothing.

The energy monitor shows whether a run behaved. In a stable model the interior energy rises as the pulse is launched, and decays to nothing once the pulse has left through the ports and absorbers (Section 4.1.4). Dissipation records the ohmic power $P = \sum \sigma |\mathbf{E}|^2 dV$. Its panel says so when no material in the document is lossy, since the series is then zero throughout.

Voltage and **Current** integrate along a drawn curve: voltage is $\int \mathbf{E} \cdot d\mathbf{l}$ along an **open** curve, current $\oint \mathbf{H} \cdot d\mathbf{l}$ around a **closed** one. Neither has a panel. Draw the sketch first and add the monitor, then drag the sketch onto the monitor in the tree to assign it.

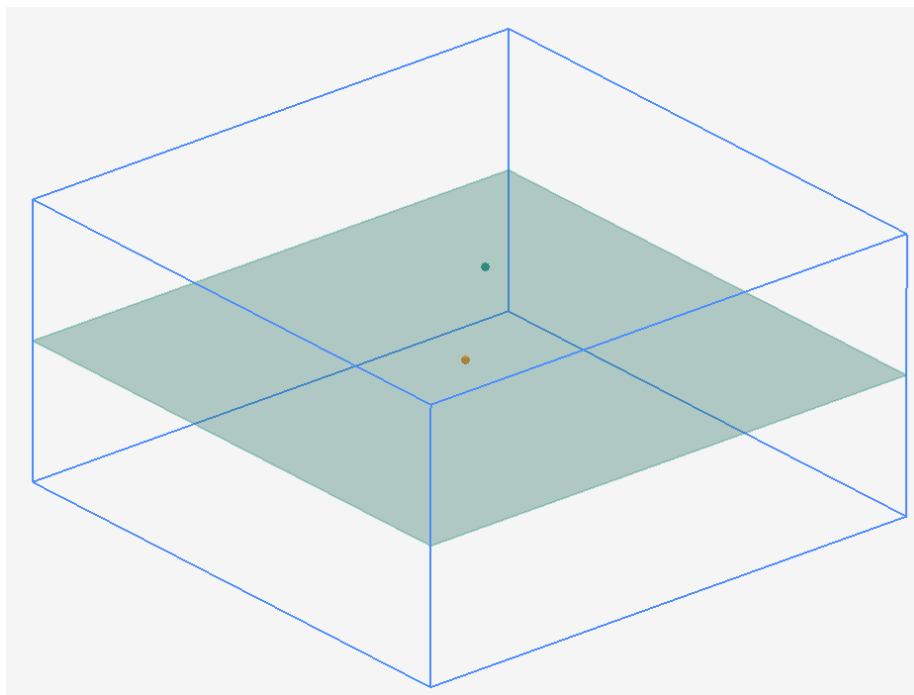


Figure 8.6: A point source (amber), a probe point and a snapshot plane (both teal) in one 3-D view, inside the domain box.

8.6 Running a simulation

Run Simulation goes through four stages, each with its own window.

The overwrite prompt appears when results for the document already exist on disk. It names the folder and offers **Overwrite** and **Run**. There is no versioning: a new run replaces the previous results, and the only way to keep them is to move or rename the folder first.

Voxelisation maps the geometry onto the grid, with a progress window and a **Cancel** button. It runs inside FreeCAD and is the slowest part of setting up a large model.

The solver shows a progress window titled **Wavesim Run**, with the stage, the step count, the elapsed time, an estimate of the time remaining and the rate in steps per second. The first run of a session may pause for a while on **Loading solver (first run may compile, please wait)...** while the solver compiles its code. This happens once per session.

FreeCAD does not respond while a run is in progress

The progress window updates and its **Cancel** works, but the rest of FreeCAD is blocked until the run finishes or is cancelled. This is expected; FreeCAD has not hung. A cancelled run has already deleted the previous results from disk, so a cancellation leaves neither the old results nor new ones.

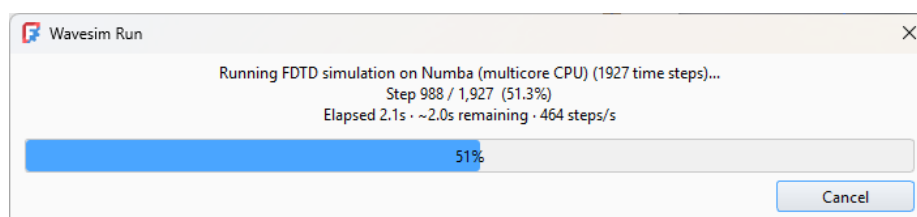


Figure 8.7: The solver progress window mid-run.

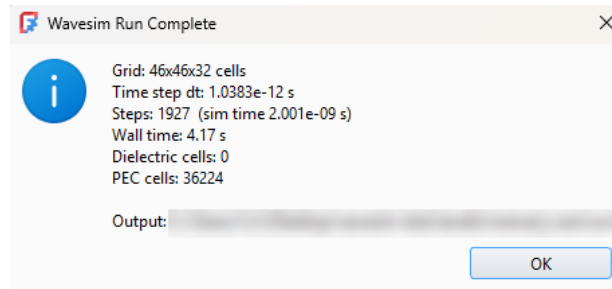


Figure 8.8: The completion summary.

The completion summary reports the grid, the time step, the steps and simulated time, the wall time, the backend, the PEC and dielectric cell counts and the output folder. A cell count or time step far from what was expected means the mesh is not the one intended, and this is the cheapest place to notice it.

The results then appear in the Results group of the tree. Chapter 10 covers reading them, and Chapter 11 what was written to disk.

Local refinement saves cells, not steps. Δt follows the smallest cell anywhere in the grid (Section 8.1), so refining one region gives a smaller model at nearly the same step count. On the coaxial line of Chapter 4, refining the conductors instead of the whole domain cut the cells by 3.7 $\times$ but the steps by only 12%; the total work fell by 4.1 $\times$, almost all of it from the cell count.

Electrostatic simulations

An electrostatic run finds the charges on conductors held at fixed potentials, and the capacitances between them. It uses the same document, geometry, materials and mesh as a full-wave run, but has no time axis, no waveform and no absorber, and finds the potential in a single solve. Section 4.2 works through a parallel-plate capacitor from start to finish.

9.1 The static problem and capacitance

THE PHYSICS · skippable

Laplace's equation, two kinds of conductor, two kinds of capacitance

With nothing changing in time, the electric field is the gradient of a potential, $E = -\nabla\varphi$, and Gauss's law in a charge-free dielectric becomes

$$\nabla \cdot (\epsilon \nabla \varphi) = 0$$

solved on the same grid as a full-wave run. A PEC body is an **equipotential**: φ is one constant throughout it. Each face of the box is either **Ground**, $\varphi = 0$ (a Dirichlet condition), or **Symmetry**, $\partial\varphi/\partial n = 0$ (a Neumann condition, meaning no field line crosses it).

A conductor has one of two conditions, never both. A **driven** conductor is held at a potential and draws whatever charge that requires; the charge is the result. A **floating** conductor is held at a fixed total charge, usually zero, and the potential it settles at is the result. An unconnected trace beside a driven one is floating; a shield tied to the chassis is driven, at 0 V.

The **Maxwell capacitance matrix** links the charges to the potentials of N conductors:

$$Q_i = \sum_j C_{ij} V_j, \quad C_{ij} = \frac{\partial Q_i}{\partial V_j}$$

with every other conductor at 0 V. This is the quantity a field solve measures. Its diagonal is positive, its off-diagonal entries negative, and each row sums to that conductor's capacitance to **ground**, here the Ground faces of the box.

The **mutual (two-terminal) capacitance** is the lumped capacitor drawn between two pins in a circuit:

$$C_{ij}^m = -C_{ij} \quad (i \neq j), \quad C_{ii}^m = \sum_j C_{ij}$$

so that the off-diagonal entries change sign and the diagonal becomes the capacitor to ground. The two matrices carry the same information, but their entries are not interchangeable: a Maxwell diagonal quoted as the capacitance of a two-plate capacitor also includes that plate's capacitance to the surroundings. The workbench reports both, under their own names.

The **field energy** is $W = \frac{1}{2} \sum_i Q_i V_i$, a check on any run.

9.2 Switching modes, and what changes

Set **Solve:** to **Electrostatic** in the Simulation panel (Section 3.1). The panel hides the time rows and the maximum frequency, which have no meaning in a static problem, and shows the checkbox **Extract the capacitance matrix**, on by default (Figure 9.1). Everything else follows from that one setting:

Table 9.1: What differs between a full-wave and an electrostatic run.

	Full wave	Electrostatic
Box faces	PML or PEC	Ground or Symmetry (Section 9.4)
Absorber	a PML layer inside the box	none
Cell size from	cells per wavelength at the max frequency	cells across the thinnest feature
Driven by	sources and ports	potentials on the PEC bodies (Section 9.3)
Records	time series, field animations, spectra	one ϕ , E or D slice per snapshot monitor, the conductor charges, the capacitance matrix
Compute backend	the Settings choice	always the CPU (Numba)

Point sources, modal ports, lumped ports and probes stay in the document but are ignored, and the run says so in the Report view. Their buttons are greyed out and **Set Potential** becomes available (Section 5.2). The energy, voltage and current monitors record nothing, since there is no time axis. A snapshot monitor's panel offers ϕ , E or D and hides its recording interval. Switching back to full wave restores all of it, because the two modes keep separate boundary settings and resolutions (Section 6.2).

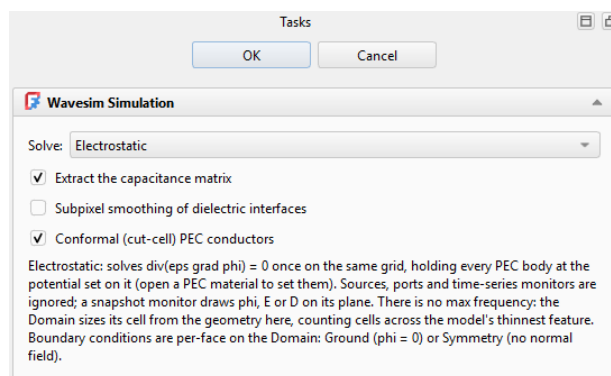


Figure 9.1: The Simulation panel in electrostatic mode.

Capacitance extraction costs one extra solve per conductor: each is driven to 1 V in turn, with the others at 0 V. The matrix is therefore independent of the potentials set on the bodies, which determine only the field shown in the slices and the charges in the conductor table. Extraction needs at least two separate conductors. With one, the run skips it and says why; that conductor's capacitance to ground is its charge at 1 V.

9.3 Conductor conditions: the Set Potential panel

Set Potential opens a panel that lists every PEC body in the document, whichever PEC material it belongs to, in four columns: **Conductor**, **Condition**, **Value** and **Held at** (Figure 9.2). Each body's **Condition** is one of three:

- **Potential** — held at the voltage in **Value**.
- **Floating** — held at the charge in **Value**, in picocoulombs (normally 0). Its potential is an output of the run.
- **Default** — no request of its own (the default for a new conductor).

A body on **Default** is never left undefined. If it touches a body that was set, it inherits that body's condition: two solids in contact are one piece of metal, with one potential. If it touches nothing that was set, it takes the document's fallback, chosen at the foot of the panel: **Conductors nobody set: is Grounded (0 V) or Floating (0 C)**. The **Held at** column shows each body's resolved condition and

where it came from. Because of this inheritance, an enclosure built from several parts needs only one of them set: that part fixes the whole lump in contact with it.

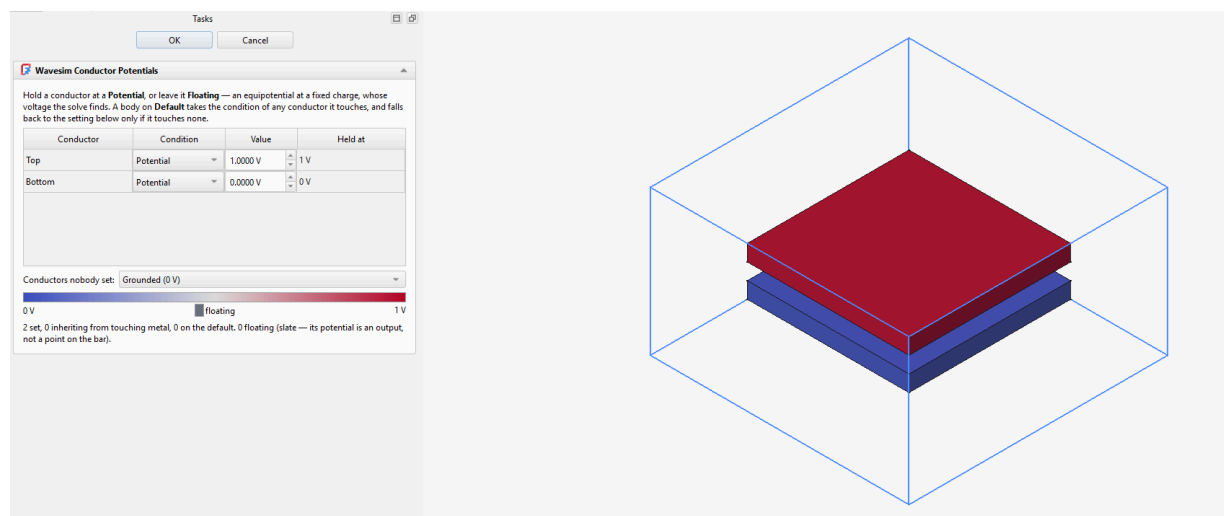


Figure 9.2: The Set Potential panel, with the 3-D view coloured by potential: Top at 1 V (red), Bottom at 0 V (blue).

While the panel is open, the 3-D view colours every conductor by its resolved potential, blue through light grey to red, with the scale beneath the table. A floating conductor is drawn in dark slate, a colour not on the scale, because its potential is an output of the run. The original colours return when the panel closes.

The panel flags three contradictions in red: touching bodies held at different potentials, touching bodies floating with different charges, and one lump both driven and floating. The row then reads **conflict**, and the solver refuses all three.

The panel creates no object. The conditions are properties of the bodies, visible in each body's property editor. Nothing is written until OK; Cancel leaves the document as it was.

9.4 Boundaries and mesh in electrostatic mode

Each box face is **Ground**, the default, or **Symmetry**. **Ground** makes the box the reference conductor against which every capacitance to ground is measured. **Symmetry** lets no field line cross the face. It halves a mirror-symmetric model, and closes a shielded structure that has no ground of its own. With every face on Symmetry, each Maxwell row sums to zero, and the capacitance window says so; the mutual capacitances remain exact.

A conductor touching a Ground face is shorted to it

A body that reaches a grounded face is electrically part of the box. The potential solve can still succeed if that body is at 0 V anyway, but capacitance extraction drives it to 1 V and fails. The run warns before it starts, names the body and the faces, and gives three remedies: set those faces to Symmetry, which suits a shielded structure and leaves the mutual capacitances exact; add background spacing; or untick **Extract the capacitance matrix**. A floating body on a Ground face cannot float at all, and the run refuses even the first solve. For an enclosure drawn up to the box edge, Symmetry is the usual remedy.

The cell size is the thinnest bounding-box dimension of any material body, conductors included, divided by **Resolution**: in cells per smallest feature, **10** by default (Section 6.4). It is derived again whenever the set of assigned bodies changes, so it does not matter whether the mode is switched before or after the materials are assigned. It also means that, in electrostatic mode, a typed cell size is replaced whenever a body is assigned or removed.

With no time step to protect, a fine cell costs only its share of the solve (Section 6.8), so resolution is cheaper here than in full wave. The graded mesh and `Refine Body Mesh` behave as in Chapter 6.

9.5 Running, and reading the capacitance matrix

`Run Simulation` works as it does for a full-wave run (Section 8.6). The completion summary lists each conductor's potential and charge, the field energy and the number of conductors in the matrix. The Results group gets one entry per snapshot monitor, and one entry named **Capacitance matrix** (**Electrostatic solution** when extraction was off or skipped) that opens the window of Figure 4.11, shown for the parallel-plate capacitor of Section 4.2.

The window has three parts. The header gives the field energy and the solver's own report: the method (`cg` for conjugate gradient), the number of unknowns and the iterations taken. **Conductors** lists each conductor, driven or floating, with its potential and charge; for a floating conductor the charge is the input and the potential the output, as a note under the table says. Below it are the two matrices of Section 9.1, each with its name and definition written above it. The **mutual** matrix gives the capacitor between two conductors; the **Maxwell** matrix is the one the field solve measures directly.

The ϕ , E and D slices open in the snapshot viewer (Section 10.4), without the Play button or frame slider: a static solve has one frame.

Results and post-processing

A completed run's output appears in the tree under **Results**, one entry per recorded quantity. Double-clicking an entry opens its window. This chapter covers the entries, the windows they open and the controls those windows carry.

10.1 The Results tree

Each monitor gives one entry, named after the monitor, and each port gives the entries its kind calls for:

- **Snapshot** — a field-slice movie (Section 10.4).
- **Energy, Dissipation, Probe, Voltage, Current** — a time series (Section 10.2).
- **Modal port** — a **group** of its own, holding the port's mode shape, its voltage and current, and the impedance $Z(f)$ computed from them. They are grouped because they are one measurement of one plane.
- **Lumped port** or **SPICE line port** — no plane and so no mode: its voltage, current and impedance entries sit directly under Results.
- **Electrostatic run** — one **Capacitance matrix** entry (Section 10.6).

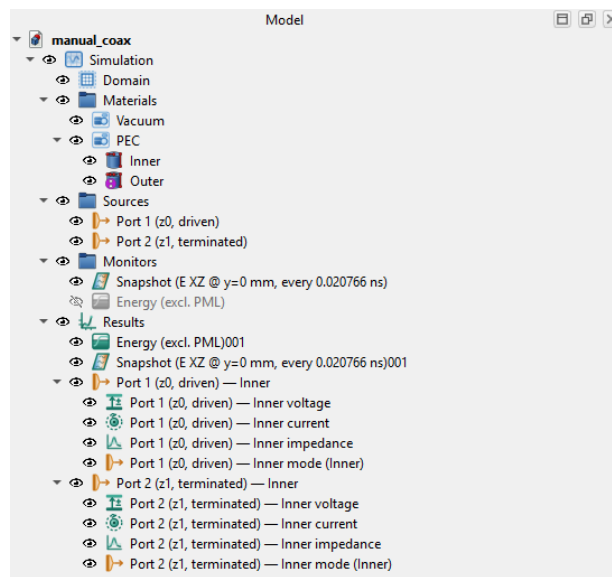


Figure 10.1: The Results group after a run of the coax example.

Run Simulation discards the whole Results group and builds a new one, so save anything worth keeping from a run — a figure, a number — before starting the next. The **Port Matrix** group (Section A.3) is separate, sits beside Results, and survives a run.

Results live in the run folder, not in the document

An entry holds only a pointer to the run folder (Section 11.2) and the name of its data there; each window reads that data when it opens. If the folder has been moved, deleted or overwritten — by another document with the same file name, or by a new run that was started and cancelled — the entry stays in the tree but opens a warning, “The result data is missing. The run output may have been moved or deleted”. The electrostatic summary is the one exception: its numbers are stored on the entry itself.

Every plot window has the standard matplotlib toolbar: home, back and forward, pan, zoom, subplot spacing, axis settings and **Save**. **Save** writes the figure as PNG, PDF or SVG at the window's current zoom and settings, and is the only way to export an image. The numbers themselves are in the run folder's `results.npz` (Appendix B).

10.2 Time-series windows

Energy, dissipation and probe entries open a single plot against time. Voltage and current entries — from a voltage or current monitor, a modal port, a lumped port or a SPICE port — add a row of two boxes beneath the plot (Figure 10.2):

- **View** switches the record between `Time domain` and three frequency views: `Spectrum  $|X(f)|$` , `Spectrum  $|X(f)|$  (dB)` and `Spectrum phase`.
- **Window** selects the taper applied before the transform: **none (as recorded)**, Tukey, Exponential, Hann or Hamming. It is greyed out in the time view.

A spectrum's frequency axis opens on the band where the signal is above one-thousandth of its peak. For a Gaussian pulse this reaches well past the meshed band: an 8 GHz pulse opens on about 30 GHz. Use the toolbar's zoom to return to the band of interest.

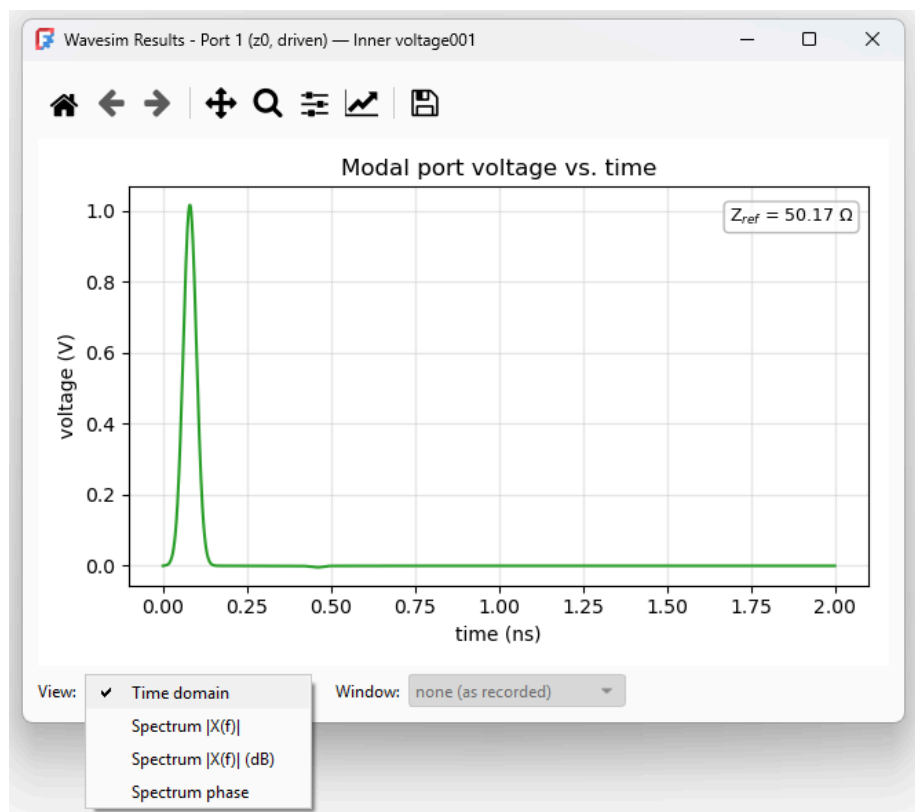


Figure 10.2: A port voltage with the View box open.

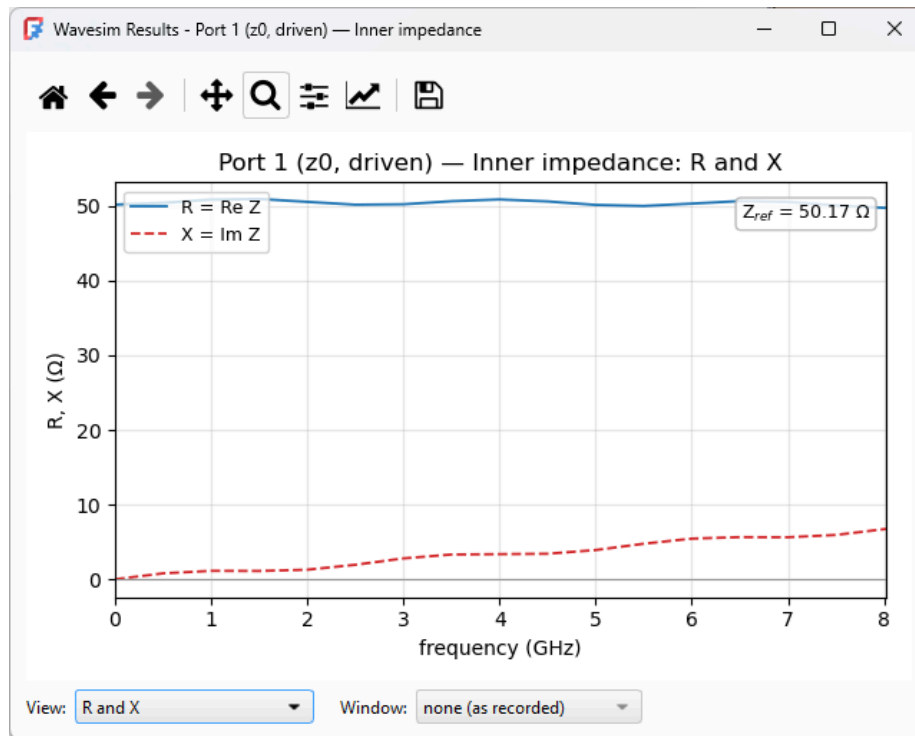


Figure 10.3: The coax impedance in the R and X view.

10.3 Frequency domain: impedance

An impedance entry divides the voltage spectrum by the current spectrum. Its View box offers two presentations. Its Window box is always available, because both series receive the same taper, which largely cancels in the ratio:

- **Bode ($|Z|$ and phase)** — magnitude above, phase below. Read a resonance and its Q here; a phase of $\pm 90^\circ$ means a purely reactive impedance.
- **R and X** — real and imaginary parts on one linear axis (Figure 10.3). The lumped content is easy to read here: a series inductance appears as the straight line $X = 2\pi fL$, and a zero crossing of X marks a resonance.

A modal port's impedance plot has a dotted guide line and a corner label at the port's **reference impedance** Z_{ref} , the Z_0 of its mode (Section 8.3), so the departure of the measured $Z(f)$ from it can be read directly. In Figure 10.3, R lies on the 50.17 Ω line across the band.

Gaps in an impedance curve are not missing data. They are frequencies where the excitation put in almost no energy, so the ratio would be one round-off error divided by another; these points are left out. If a plot is mostly gaps, the spectrum view of the voltage or current entry shows the band the drive actually covered.

10.4 The snapshot viewer

A snapshot entry opens the viewer of Figure 10.4: one colour map of the field on the monitor's plane, in model coordinates, with a row of controls beneath. A control with nothing to act on is left out: **Mask PEC** appears only where the slice contains conductors, **Mask PML** only with an absorber, **Geometry** only where there are outlines to draw, and **Vectors** only for a vector field.

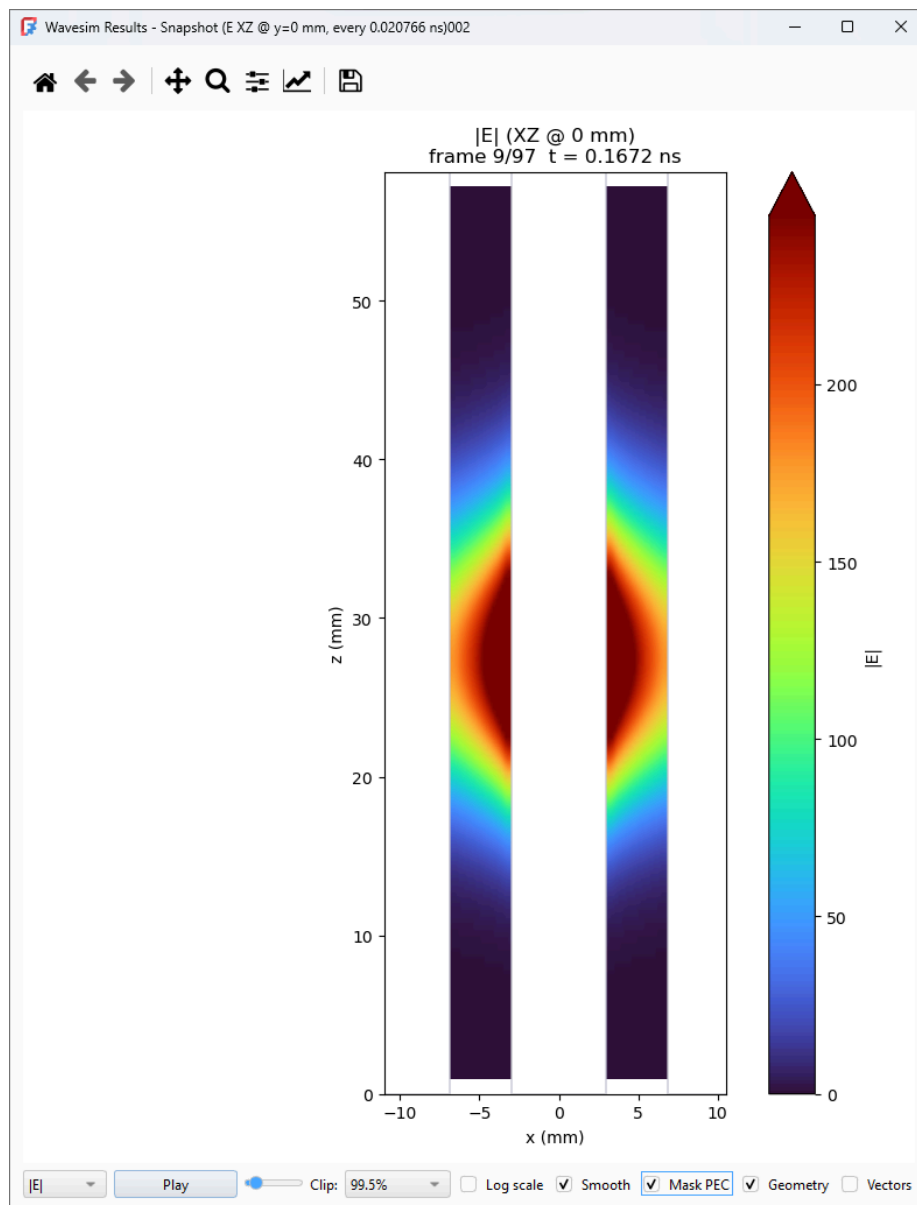


Figure 10.4: The snapshot viewer on the coax's E-field slice.

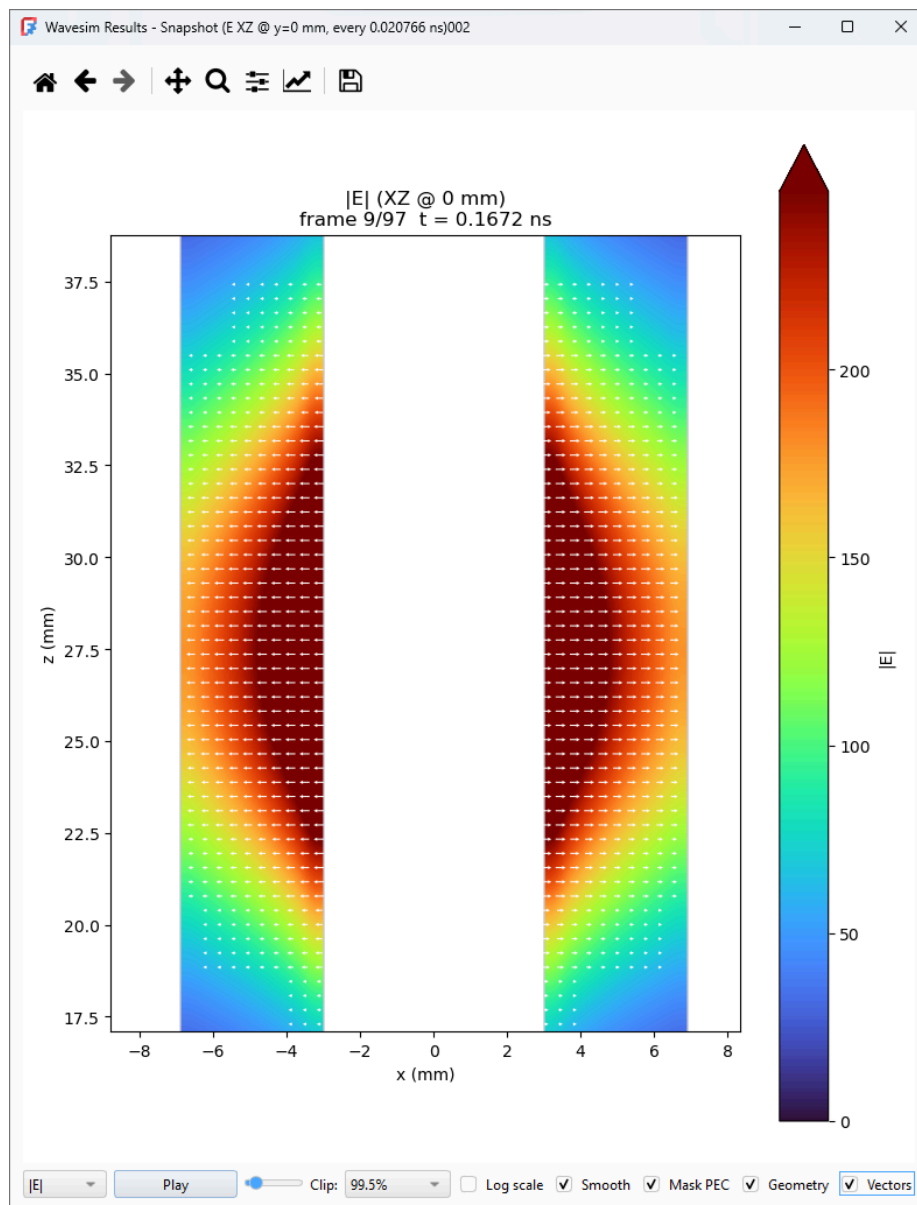


Figure 10.5: The same frame zoomed in, with Vectors on.

Table 10.1: The snapshot viewer’s controls.

Control	What it does — and when to change it
Component	<code>Ex</code> , <code>Ey</code> , <code>Ez</code> or the magnitude <code> E </code> , the default. Signed components use a blue–red map symmetric about zero, magnitudes a sequential one. Each is scaled on its own peak, so a weak component is not washed out.
Play, slider	Animate or step through the frames. Hidden for an electrostatic slice, which has only one.
Clip	The percentile of the field mapped to the top of the colour scale: 99.5% , 99.9, 99, 98, 95 or <code>peak</code> . Below <code>peak</code> , the few hottest cells (a source point, a conductor edge) saturate, so the rest of the slice uses the full colour range. A lower value helps a slice that shows almost one colour; <code>peak</code> shows the true maximum.
Log scale	A logarithmic colour scale, for fields spanning decades — decay into an absorber, leakage through a gap. Clip greys out: a log scale needs no clipping.
Smooth	Shades between neighbouring cells instead of painting each cell flat; it adds no data. Switched off, it shows the actual cells and speeds up playback on a fine grid.
Mask PEC	On by default, with two effects: it blanks the metal, cut at the CAD surface rather than the staircase, and it keeps the zeros inside the metal out of the smoothing, which would otherwise paint a false dark fringe along every conductor. Switched off, it shows what the run holds there.
Mask PML	On by default. It blanks the absorbing layer, where the field is absorbed by design. The dashed line marking the start of the absorber is drawn either way, and with the mask off a wave can be seen decaying into it.
Geometry	On by default. It outlines each material’s cross-section in its own colour. The outlines are cut from the current CAD model, not from the voxels, so they show the true shape rather than the staircase; a body moved since the run is drawn where it is now.
Vectors	Overlays the in-plane field as arrows (Figure 10.5). Arrow length follows the field up to a cap, and arrows too short to show a direction are left out, so weak regions fade to plain colour rather than filling with dots.

10.5 Mode results

A modal port’s mode entry opens the transverse mode shape the port launched and absorbed (Figure 10.6): the potential ϕ across the port plane in colour, the transverse E field as arrows, and the conductors outlined. The box in the upper left lists the energised conductor, the plane’s position, Z_0 , ϵ_{eff} , the per-unit-length C and L , the phase velocity and the band. These are the quantities `Compute Mode` reports before a run (Section 8.3), stored with the result. The window has **Smooth** and **Mask PEC** controls, which behave as in the snapshot viewer.

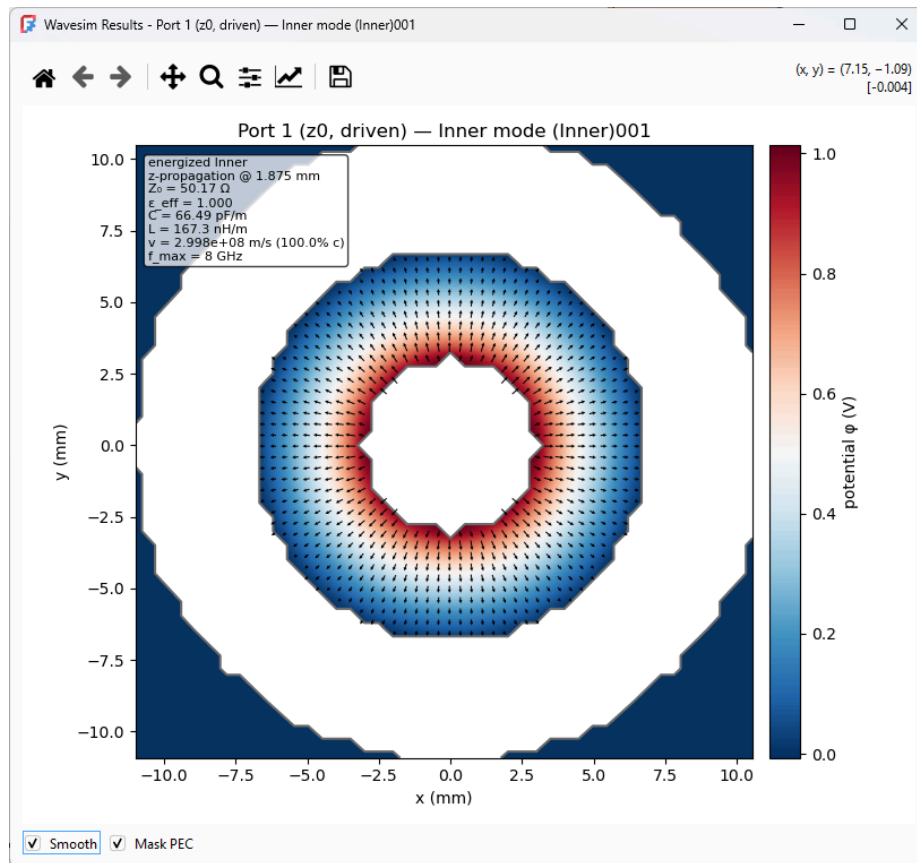


Figure 10.6: A stored mode result: the coax's TEM mode, with its parameter box.

10.6 Electrostatic results

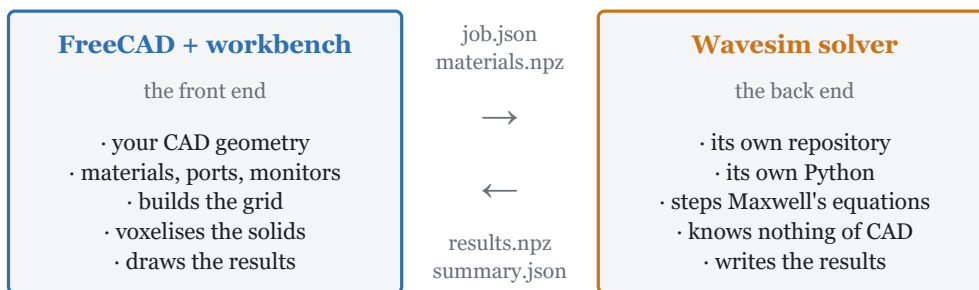
An electrostatic run adds one entry per snapshot monitor, each a single ϕ , E or D frame in the viewer above, with no Play button or slider. It also adds one **Capacitance matrix** entry, named **Electrostatic solution** when no matrix was extracted. Its window (Section 9.5) lists each conductor's condition, potential and charge, then the Maxwell and mutual capacitance matrices, each labelled with its convention and printed in automatic units (fF, pF, ...). It is a plain table window, with no plot toolbar and no export; the numbers are copied by hand. It is also the one result stored in the document, so it still opens after the run folder is gone.

How the workbench uses the solver

11.1 Two programs, one document

The workbench and the Wavesim solver are separate programs, running in separate Pythons (Section 2.1). They share files on disk and nothing else. When `Run Simulation` is pressed:

1. The **workbench** converts the document into numbers. It builds the grid, cuts every material body into cells (voxelisation) and writes the job and the material arrays into the run folder. This happens inside FreeCAD, behind a `Voxelizing geometry...` progress window.
2. It starts the **solver** as a separate process: the solver's Python, running the workbench's `runner.py` on that folder.
3. The solver reads the folder, runs the simulation, and writes its results back into it.
4. The workbench reads the results and builds the Results group in the tree.



Two separate programs. They share files on disk and nothing else.

Figure 11.1: The front end and the back end. Only the four files cross between them.

FreeCAD does not respond while the solver runs (Section 8.6). The progress window stays live: its first line reports the solver's current stage, and during a full-wave run it names the backend in use. `Cancel` stops the solver immediately.

If the run fails, an error box shows `The solver run failed:` followed by the solver's own message, and the Report view repeats it.

11.2 What is written to disk

Each document has its own folder inside the results folder set in Settings (Section 2.3):

```
<results folder>\<document name>\run\
```

Table 11.1: The files of a run folder.

File	Written by	Contents
<code>job.json</code>	workbench	The grid, the boundaries, the sources, ports and monitors, the run length, and the solver settings.
<code>materials.npz</code>	workbench	The material of every cell.
<code>results.npz</code>	solver	Everything the monitors and ports recorded.
<code>summary.json</code>	solver	Grid size, time step, step count, run time, the backend used — or the error, if the run failed.

Appendix B describes the four files in enough detail to open them in Python.

Name	Type	Size
 job.json	JSON Source File	13 KB
 materials.npz	NPZ File	119,252 KB
 results.npz	NPZ File	457 KB
 summary.json	JSON Source File	2 KB

Figure 11.2: A run folder.

Two consequences follow.

- **Each run replaces the last.** The folder name depends only on the document, so a new run deletes the four files before it starts — after asking, if results exist (Section 5.4). To compare two variants, save each as its own document, or copy the run folder aside first. A cancelled run has already deleted the old results. The Results group still lists them, but its entries no longer open (Section 10.1).
- **The folder is named after the file, not its location.** A saved document uses its file name without the extension, so two documents with the same file name in different folders share one run folder. An unsaved document uses its label, such as `Unnamed`. Save a document before running it.

Other folders beside `run` hold other jobs. A `Port Matrix (Z / Y)` sweep writes to `matrix`, with one subfolder per run, `drive_0`, `drive_1` and so on; it never touches `run`, and an ordinary run never touches `matrix`. `Compute Mode` on a modal port uses a temporary folder, deleted once the mode has been displayed.

11.3 Compute backends

vo.1 has no GPU support

Simulations run on the CPU, across all its cores. Choose `Numba (multicore CPU)`, or leave `Automatic (CUDA GPU if available, else Numba)` on a machine without an NVIDIA GPU.

The Settings dialog lists four backends. In vo.1 they behave as follows:

Table 11.2: The solver backends offered in Settings.

Setting	What happens
Automatic	Selects Numba, the multicore CPU backend, for any run with conformal PEC or a lossy material. That is the usual case, since conformal PEC is on in every new simulation. On a machine with an NVIDIA GPU, a run with neither goes to the GPU, which vo.1 does not support; on such a machine, choose Numba.
CUDA GPU	Not supported. Any run with conformal PEC or a lossy material is refused with an error.
Numba	The supported backend.
NumPy	Much slower; kept as a reference implementation.

Electrostatic runs, and the mode solve behind `Compute Mode`, always run on the CPU, whatever the setting.

The `Voxelisation` setting controls something else: the number of processes the workbench uses to cut the geometry into cells before the solver starts. The default is one per core, less one. `Serial (one process)` produces exactly the same cells, more slowly.

11.4 Re-running a stored job

A run folder holds everything the solver needs, so a run can be repeated from a terminal, without FreeCAD:

```
<your solver environment>\python.exe <workbench folder>\runner.py <run folder>
```

`<workbench folder>` is the installed `wavesim-workbench` folder (Chapter 2), and `<run folder>` is, for example, `<results folder>\<document name>\run`. The solver prints its progress as `PROGRESS` and `STATUS` lines and ends with a line beginning `DONE`, replacing `results.npz` and `summary.json` in that folder.

This suits a long run: FreeCAD stays free throughout, and can even be closed. The `"backend"` entry in `job.json` can also be changed before re-running. `job.json` names the Wavesim repository by its full path, so on another machine, update `"wavesim_path"` to match.

The workbench does not rebuild the Results group after a re-run. Most entries read `results.npz` when they open (Section 10.1), so they show the re-run's data; the electrostatic summary is stored on its entry and keeps the old numbers. Anything else is read in Python (Appendix B).

Advanced features

These features are experimental. They work, but they are less exercised than the rest of the workbench. Check their results against a case whose answer is already known before relying on them.

A.1 Gaussian beam sources

A **Gaussian beam** launches a beam with a Gaussian transverse profile from one domain face into the domain. It needs no geometry on the face, which makes it the source for illuminating a scatterer, an aperture or a lens with a known polarisation.

Add Gaussian Beam creates the beam on the Z min face and opens its panel. The panel has four rows of its own, followed by the waveform rows of Section 8.2:

- **Launch face:** — one of the six faces, each labelled with the direction the beam travels, such as Z min (z0) → +Z.
- **Polarization angle:** — the direction of E within the face, in degrees (default **0**).
- **Beam waist w0:** — the waist, in millimetres (default **auto**).
- **Directional (one-way launch into the domain)** — **on**. Off, the source is a bare E sheet that radiates both ways, half of it into the absorber behind.

On OK the launch face is set to PML, and the Domain panel shows it as Port / beam (absorbing) (Section 6.2). The beam starts from a plane one absorber depth inside the face, and the absorber behind it takes the backward radiation.

Polarisation. The angle is measured in the face's own transverse frame, from its first axis towards its second; with the inward normal the two axes form a right-handed set. The panel names both axes for the selected face; on Z min, 0° points E along $+x$ and 90° along $+y$. The same physical polarisation therefore takes a different angle on opposite faces: E along z is 90° on X min but 0° on X max.

Table A.1: The transverse axes of each launch face (angle 0°, then 90°).

X min	X max	Y min	Y max	Z min	Z max
y, z	z, y	z, x	x, z	x, y	y, x

Waist. w_0 is the radius at which the E amplitude falls to $1/e$ (intensity to $1/e^2$). The beam starts with a flat phase front, so the waist sits on the launch plane. It stays collimated over the Rayleigh range $z_R = \pi w_0^2 / \lambda$ and diverges beyond it at $\theta \approx \lambda / (\pi w_0)$; a waist many wavelengths across approximates a plane wave. A waist of 0, shown as **auto** (half the face half-width), takes half of the smaller half-width of the face inside the absorber. Beneath the rows, the panel reports the resolved w_0, z_R at the waveform's frequency, and the fraction of the on-axis amplitude still present at the edge of the absorber.

The beam must fit inside the face

The launch sheet is set to zero over the absorber cells at the edges of the face. A beam that still carries amplitude there is cut off sharply, and diffracts from the cut. At the automatic waist the amplitude at the edge is $e^{-4} \approx 1.8\%$. Where a wider beam is needed, widen the domain rather than enlarge the waist, and check the panel's edge figure.

The launched amplitude is proportional to the waveform's `Amplitude`, but is not calibrated to a field strength. Measure the level with a probe where an absolute value is needed.

A.2 SPICE co-simulation

A SPICE port couples the full-wave run to a circuit in ngspice. The two advance together: at each time step the circuit sees the port as a voltage source, taken from the fields, behind a small series resistance, and returns the current that flows, which the port injects into the grid. This suits a nonlinear or many-component circuit. A plain R, L or C belongs in a lumped port (Section 8.4), which needs no ngspice. SPICE ports need `PySpice` and the ngspice shared library, whose path is set in Settings (Section 2.3).

There are two ways to attach a circuit:

- **SPICE line port.** Add SPICE Line Port creates a straight port across a gap, like a lumped port. Its line comes from a sketch: draw a two-point sketch and drag it onto the port in the tree, or choose it under `Port curve:`. The start of the sketch is the `+` terminal. A port without a sketch is skipped at run time, with a message in the Report view.
- **SPICE drive of a modal port.** In the Modal Port panel, set `Drive:` to `External SPICE circuit`. The circuit then drives the mode of the cross-section. Unlike the waveform drive, the launch is from a plane just inside the face, and the face is set to `PML (Port / beam (absorbing))`, so carry the line straight out through the face (Section 6.3). The conductor table is hidden, and `Inject fields:` chooses **E and H (directional)** or `E only (bidirectional)`.

Both take the same three rows: `Netlist:`, with a `Browse...` button; `Node + :`, default **port1p** and `Node - :`, default **0**. The netlist must follow four rules:

1. **Both nodes already exist in the netlist.** The workbench connects its own port element between them; add no source or port element for the port.
2. **Each node has a DC path to ground node 0.** Using `0` as the minus node is the simplest way.
3. **The workbench runs the analysis.** Analysis and output cards (`.tran`, `.ac`, `.op`, `.print`, `.meas`, `.end` and the like) are removed, and a transient stepped at the FDTD time step is added. The first line is the title, as in any SPICE deck.
4. **One netlist drives one port.** Several SPICE ports run separate circuits, not one shared circuit.

A netlist that satisfies all four, for a 1 V pulse generator with 50 Ω source resistance and a diode limiter across the port:

```
* Pulse generator with a diode limiter across the port
V1 src 0 PULSE(0 1 0 50p 50p 1n 2n)
R1 src port1p 50
D1 port1p 0 DLIM
.model DLIM D(IS=1e-14 N=1)
.tran 1p 5n
.end
```

Node `port1p` and ground exist, and `port1p` reaches ground through `R1` and `V1`. The `.tran` and `.end` lines are removed at run time, so the same file still runs on its own in ngspice. Each SPICE port records its voltage and current (Section 10.1). Two properties appear only in the property editor, both **off**: `UseInitialConditions` starts the transient from the netlist's initial conditions instead of its DC solution, and `InvertPortCurrent` reverses the sign of the port current.

A.3 The port matrix (Z / Y)

A single run gives each port's impedance with every other port terminated as it is, so the answer changes when a termination changes. The **impedance matrix** Z , defined by $V = ZI$ over the port voltages and currents, does not: it is a property of the structure, from which the response to any set of terminations follows.

Each run supplies one voltage and one current per port: N equations against the N^2 entries of Z . An N -port matrix therefore needs N independent excitations, and **Port Matrix (Z / Y)** runs the solver once per selected port, driving that port while every other source in the document is silenced. The geometry is voxelised once for the whole sweep; only the solve repeats.

Waveform-driven modal ports, one entry per conductor row, and lumped ports can take part. The command opens the **Wavesim: Port Matrix** dialog, which lists them with every one ticked, states the size and cost (N ports $\rightarrow$ an $N \times N$ matrix, N solver runs.), and starts with **Run sweep**. The overwrite question follows (Section 5.4), and each run shows its own progress window, **Drive k of N: ...**.

Ports left out stay in their terminations and become part of the network the matrix describes. A 2×2 block of a six-port structure costs two runs, and gives the two-port matrix of the structure **including** the other four terminations, which is usually the quantity wanted. Every run presents the same terminations: a lumped port with no drive is given a voltage source in series with its load, which at zero amplitude leaves the load unchanged. All runs use one waveform, that of the first selected port that carries one.

The result is a **Port Matrix** group beside Results, written to the **matrix** folder (Section 11.2). It holds **Z matrix** ($N \times N$) and one group per run, **Drive k: ...**, with every port's voltage and current in that run. The matrix window draws each entry in an $N \times N$ grid on one shared scale, with these controls:

- **View — R and X**, Magnitude $|Z|$ (log), Equivalent L (Im/ω) or Equivalent C ($-1/\omega \cdot \text{Im}$). An entry whose equivalent L or C is flat across the band behaves as a lumped element there.
- **Quantity** — Z (impedance) or $Y = Z^{-1}$ (admittance), the same measurement inverted frequency by frequency. A shunt element is easier to read in Y .
- **Window** — the tapers of Section 10.2, applied to every record of the sweep.
- **Read at** — a frequency slider, marked on every plot, with the matrix at that frequency in a table beneath.

Under the grid, a **quality strip** plots two curves. The solid one is the reciprocity error $\max |Z_{ij} - Z_{ji}| / \max |Z|$, which is small for a passive, reciprocal structure. The dashed one is the conditioning of the N drives, 1 being ideal. Frequencies where the conditioning exceeds 10^{10} , or where a drive put in almost no energy, are not reported, and appear as gaps. A band where either curve climbs is a band where the runs did not measure one linear network, and the matrix there is not to be trusted.

What the matrix is not

The matrix is reported **at the port planes** and is not de-embedded: it includes any line between a plane and the component, and a lumped port's own cell capacitance (Section 8.4). No equivalent circuit, T or Π , is fitted to it. It is not an S -parameter matrix, and no Touchstone file is written. A document that contains a SPICE port is refused, because a circuit-driven source cannot be silenced between runs.

Files a run writes

A run folder (Section 11.2) holds two JSON files, readable in any text editor, and two NumPy archives. All four are in SI units — metres, seconds, volts, amperes, joules — and every coordinate is measured in metres from the minimum corner of the domain, not from FreeCAD’s origin. They open in the solver’s Python:

```
import json
import numpy as np
run = r"<results folder>\<document name>\run"
summary = json.load(open(run + r"\summary.json"))
data = np.load(run + r"\results.npz")
print(data.files)           # every array in the archive
t = data["probe_0_times"]    # seconds
v = data["probe_0_values"]
```

job.json is the job as the workbench wrote it: the `grid`, the `boundary`, one list per kind of source and port, the `monitors` and the step count, with an `electrostatic` block for an electrostatic run. `backend` and `wavesim_path` are the entries worth editing by hand (Section 11.4).

materials.npz holds one $N_x \times N_y \times N_z$ array per quantity and field component: `eps_x`, `eps_y`, `eps_z`, `mu_x`, `mu_y`, `mu_z` and `pec_mask`. A conformal run adds six open-fraction arrays (`pec_edge_open_*`, `pec_face_open_*`), a lossy run `sigma_x/y/z`, and an electrostatic run `pec_id`, one label per conductor.

results.npz holds every recorded series as a pair of arrays, `__times` and `__values`. The index k counts objects of one kind in the order of `job.json`; their names are in `summary.json` under `probes`, `voltages`, `currents`, `ports`, `lumped_ports`, `spice_ports` and `snapshots`.

Table B.1: The arrays in `results.npz`.

Recorded by	Arrays
Energy, dissipation	<code>energy_*</code> (whole domain), <code>energy_interior_*</code> ; <code>dissipation_*</code> , <code>dissipation_interior_*</code>
Probe, path monitor	<code>probe_k_*</code> ; <code>voltage_k_*</code> , <code>current_k_*</code>
Modal, lumped, SPICE port	<code>port_kv_*</code> and <code>port_ki_*</code> ; <code>lumped_kv_*</code> , <code>lumped_ki_*</code> ; <code>spice_kv_*</code> , <code>spice_ki_*</code>
Modal port mode	<code>mode_k_m_phi</code> , <code>mode_k_m_E_Ex</code> , ... with node coordinates <code>mode_k_m_ca</code> , <code>mode_k_m_cb</code>
Snapshot	<code>snapshot_k_times</code> ; <code>snapshot_k_<comp>_data</code> per recorded component (no magnitude); <code>snapshot_k_edges0/1</code> , one entry longer than a frame, for <code>pcolormesh</code>
Electrostatic run	the slices, stored as one-frame snapshots; <code>capacitance_maxwell</code>

summary.json opens with `ok`. A failed run holds only `ok: false` and the `error`. A completed run adds `dt`, `steps`, `sim_time_s`, `wall_time_s`, `Nx`, `Ny`, `Nz`, `conformal_pec`, the cell counts and one entry per monitor; a full-wave run also records `backend`, `pml_faces`, `pec_faces`, `lossy` and one entry per port. Each `ports` entry carries the port’s `reference_impedance`, and each `modes` entry the mode’s parameters. An electrostatic run carries an `electrostatic` block with the `potentials`, `charges` and `energy`, and a `capacitance` block holding the conductor `names` and the `maxwell` and `mutual` matrices.